MIL-B-857A(SHIPS)
1 June 1956
SUPERSEDING
MIL-B-857(SHIPS)
15 March 1950

MILITARY SPECIFICATION

BOLTS, NUTS, AND STUDS

1. SCOPE

1.1 Scope. - This specification covers standard bolts, nuts, and studs for general use within the range of sizes specified hereinafter.

1.2 Classification. - Bolts, nuts and studs shall be of the following types and styles, as specified (see 6.2):

Types:
- Bolts
  - Type I - Regular hexagon head (unfinished).
  - Type II - Regular semifinished hexagon head.
  - Type III - Regular finished hexagon head.
  - Type IV - Round head, unfinished (button head).
  - Type V - Round head square neck, unfinished (carriage bolt).
- Nuts
  - Type I - Regular hexagon (unfinished).
  - Type II - Regular semifinished hexagon.
  - Type III - Regular finished hexagon.
  - Type IV - Regular semifinished hexagon jam.
  - Type V - Regular finished hexagon jam.
  - Type VI - Regular semifinished hexagon slotted.
  - Type VII - Regular finished hexagon slotted.
- Studs
  - Type I - Full body.
  - Type II - Reduced body.

Styles:
- Nuts
  - Style a - Washer faced.
  - Style b - Double chamfered.
- Studs

Style number	Tap end thread	Nut end thread
a	NC 5	UNC 3
b	NC 5	UNC 2

1.2.1 Grades. - Bolts, nuts and studs made of carbon and alloy steels shall be of the following grades, as specified (see 6.2):

Grade 1.
Grade 2.
Grade 5.
Grade 8.

FED. SUP. CLASS
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2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
QQ-M-151 - Metals; General Specification for Inspection of.
QQ-N-281 - Nickel-Copper-Alloy; Forgings, Rods, Shapes, Sheets, Strips and Wire.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-763 - Steel, Corrosion-Resisting: Bars and Forgings (Except for Reforging).
QQ-Z-325 - Zinc Plating (Electrodeposited).
LLL-B-631 - Boxes; Fiber Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-676 - Boxes, Set-Up, Paperboard.

MILITARY

JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes; Wood Nailed.
MIL-P-116 - Preservation, Methods of.
MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).
MIL-B-4229 - Boxes; Paperboard, Metal-Stayed.
MIL-L-10547 - Liners, Case, Waterproof.
MIL-C-16173 - Corrosion Preventive, Solvent Cut-Back, Cold-Application.
MIL-W-16232 - Coatings, Phosphate, Heavy (Manganese and Zinc Type) and Phosphate Treating Solutions.
MIL-C-17516 - Copper-Silicon Alloy Bars, Plates, Rods, Sheets, Strips, Flat Wire, Wire, Forgings, and Structural and Special Shaped Sections.
MIL-Z-17871 - Zinc-Coating (Hot-Dip Galvanizing).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

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NATIONAL BUREAU OF STANDARDS PUBLICATION
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials. - Bolts, nuts and studs shall be of carbon and alloy steels, corrosion resisting steels, brass, silicon bronze, nickel copper, or other material, as specified (see 6.2).

3.1.1 Carbon and alloy steels. -

3.1.1.1 Bolts and studs. - Bolts and studs made of carbon and alloy steels shall be as specified in table I for the applicable grade.

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3.1.1.2 Nuts.- Nuts made of carbon and alloy steels shall be as specified in table I for the applicable grade. Nuts of a specified grade are satisfactory for use with bolts and studs of the same or lower grades. (For example, grade 5 nuts could be used with grades 1, 2, and 5 bolts and studs.)

Table I - Mechanical and chemical properties of carbon and alloy steel bolts, nuts and studs.

Grade	Nominal size	Elastic-proof load (min.)	Tensile strength (min.)	Hardness				Chemical composition			Bend test
				Brinell		Rockwell		Carbon	Phosphorus	Sulfur	
	Inches	P.s.i.	P.s.i.	Min.	Max.	Min.	Max.	Max.	Max.	Max.	
1	All sizes	Not required	55,000	---	207	---	B95	--	05	06	180° over a mandrel having a diameter 1/2 the nominal diameter of the bolt for sizes to 1 inch inclusive. For sizes over 1 inch the mandrel diameter shall be the same as the nominal diameter of the bolt.
2	To 1/2 (incl.)	55,000	69,000	---	241	---	B100	30	04	05	
	Over 1/2 to 3/4 (incl.)	52,000	64,000	---	241	---	B100	30	04	05	
	Over 3/4	28,000	55,000	---	207	---	B95	--	04	05	
5	To 3/4 (incl.)	85,000	120,000	241	302	C23	C32	55	04	05	
	Over 3/4 to 1 (incl.)	78,000	115,000	235	302	C22	C32	55	04	05	
	Over 1 to 1-1/2 (incl.)	74,000	105,000	223	285	C19	C30	55	04	05	
8	To 1-1/2 (incl.)	120,000	150,000	302	352	C32	C38	--	04	05	

3.1.1.2.1 Nuts made of resulfurized open hearth steels in lieu of plain carbon steels are acceptable for grades 1 and 2 providing all other requirements of the specification are met.

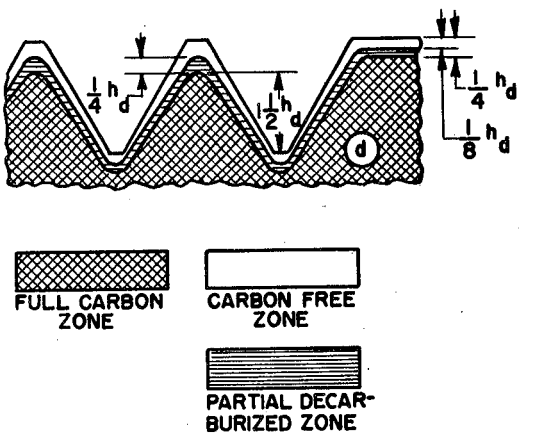
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3.1.1.3 Heat treatment. - When bolts, nuts and studs, grades 5 and 8 are heat treated, the treatment shall consist of uniformly heating from a temperature below the critical range to a temperature above the critical range to refine the structure, quenching under substantially uniform conditions, and reheating to a temperature sufficient to develop the mechanical properties, but in no case less than 800°F.

3.1.1.4 Decarburization. - The complete and partial decarburization of the threads of grades 5 and 8 bolts and studs, when tested in accordance with 4.5.1 shall not exceed the amounts specified in table II, where h_d is equal to 1/2 of the basic thread depth. It is desirable that the carbon content throughout the threads be essentially the same as that of the core.

Table II - Decarburization classification of screw threads.

T. p. i.	h_d	1.5 h_d	0.5 h_d
80	0.00406	0.00609	0.00203
72	.00451	.00677	.00226
64	.00507	.00761	.00254
56	.00580	.00870	.00290
48	.00677	.01015	.00338
44	.00738	.01107	.00369
40	.00812	.01218	.00406
36	.00902	.01353	.00451
32	.01015	.01522	.00508
28	.01160	.01740	.00580
24	.01353	.02030	.00677
20	.01624	.02436	.00812
18	.01804	.02706	.00902
16	.02030	.03045	.01015
14	.02320	.03480	.01160
13	.02498	.03747	.01249
12	.02706	.04059	.01353
11	.02952	.04429	.01476
10	.03248	.04871	.01624
9	.03608	.05413	.01804
8	.04059	.06089	.02030
7	.04639	.06959	.02320
6	.05413	.08119	.02706



SH 703

h_d = 1/2 BASIC THREAD DEPTH.

NOTE: The dimensioned zones shown on the shank merely indicate the approximate depth of bar decarburization which will produce the corresponding amount of thread decarburization when the threads are rolled. The dimensions at the shank are not part of the specification.

3.1.2 Corrosion resisting steels. - Bolts, nuts and studs of corrosion resistance steel shall be made of steel conforming to Specification QQ-S-763, condition A or C, class FS 302, 303 or 304 as specified (see 6.2). The minimum breaking load shall be that calculated from the minimum tensile strength required for the material and the cross sectional area based on the following stress area formula (see tables III, IV, XX and XXI):

$$3.1416 \left[\frac{E}{2} - \frac{3H}{16} \right]^{2}$$

Where:

E = Basic pitch diameter.

H = Depth of sharp V-thread.

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3.1.2.1 Unless otherwise specified in the contract or order, corrosion resisting steel bolts, studs and nuts shall be passivated after completion by immersion for 20 minutes in one of the following solutions:

(a) A solution containing 15 percent by weight of nitric acid at room temperature.
(b) A solution containing 6 to 8 percent by weight of nitric acid plus 1 to 2 percent by weight of hydrofluoric acid heated to a temperature of 120 to 130°F.
(c) A solution containing 50 to 70 percent by volume of nitric acid (sp. gr. 1.42) at 120 to 130°F.
(d) A solution of 20 percent by volume of nitric acid (sp. gr. 1.42) and 2 percent sodium dichromate at 120 to 130°F.

Each immersion shall be followed by a thorough rinse in hot water.

3.1.3 Nonferrous bolts, nuts and studs. - Nonferrous bolts, studs and nuts other than those specified in 3.1.3.3 and 3.1.3.4 shall be in accordance with the chemical and mechanical strength requirements as specified (see 6.2).

3.1.3.1 Heat treatment. - Cold worked brass and bronze bolts, studs and nuts shall be stress relieved by heating the parts to 600°F. and holding at that temperature for a minimum of one hour per inch of diameter or thickness.

3.1.3.2 Breaking load. - The minimum breaking load of nonferrous bolts, studs and nuts shall be that calculated from the minimum tensile strength required for the material and the cross sectional area given by the following formula:

$$3.1416 \left[\frac{E}{2} - \frac{3H}{16} \right]^2$$

Where:
E = Basic pitch diameter.
H = Depth of sharp V-thread.

3.1.3.3 Nickel copper alloy. - Monel bolts, nuts and studs shall be made from material conforming to class A of Specification QQ-N-281.

3.1.3.4 Silicon bronze. - Copper silicon bolts, nuts, and studs shall be made of silicon bronze conforming to Specification MIL-C-17516. Composition 2 material shall be used when the bolts, nuts and studs are cold headed; composition 1 when hot headed, or machined from bar stock.

3.1.4 Strength and ductility. -

3.1.4.1 Bolts and studs. -

3.1.4.1.1 Tensile strength 10° wedge. - Carbon and alloy steel, and nonferrous bolts, types I, II and III shall conform to the minimum tensile strength requirements specified in table I for the applicable grade, or as in the applicable materials specification when tested in accordance with 4.4.3.4.

3.1.4.1.2 Axial tensile strength. - Bolts, types IV and V made of ferrous and nonferrous metals, and studs types I and II of ferrous and nonferrous metals shall conform to the minimum tensile strength specified in table I for the applicable grade, or as in the applicable materials specification when tested in accordance with 4.4.4.

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3.1.4.1.3 Elastic proof load. - Carbon and alloy steel bolts and studs shall conform to the minimum elastic proof load requirements specified in table I, as applicable, when tested in accordance with 4.4.3.3.

3.1.4.1.4 Yield strength. - Nonferrous bolts and studs shall conform to the minimum yield strength requirements specified in the applicable materials specification when tested in accordance with 4.4.5.

3.1.4.1.5 Hardness. - Carbon and alloy steel bolts and studs of insufficient size for tensile test shall conform to the hardness requirements specified in table I for the applicable grade when tested in accordance with 4.4.2.

3.1.4.1.6 Bending. - Carbon and alloy steel bolts and studs, grades 1, 2, and 5, when the unthreaded portion is of sufficient length shall withstand being bent cold, 180 degrees around a mandrel, without visual evidence of fracture when tested as specified in 4.4.8.

3.1.4.1.7 Flattening. - When the bend test specified in 3.1.4.1.6 cannot be made, the bolts and studs shall withstand cold flattening along the longitudinal axis, to a thickness equal to one half the original nominal diameter when tested as specified in 4.4.9.

3.1.4.1.7.1 Bolts and studs that are case hardened, or made of steel grade 8, or made from corrosion resisting steels or nonferrous metals shall not be subjected to the bending or flattening tests specified in 3.1.4.1.6 or 3.1.4.1.7 unless specified (see 6.2).

3.1.4.1.8 Shear strength. - When specified (see 6.2), bolts and studs shall meet minimum transverse shear strength levels, when tested in accordance with 4.4.7. Minimum shear strength levels shall be as specified (see 6.2).

3.1.4.1.9 Plastic deformation. - When specified (see 6.2) bolts and studs shall conform to the plastic deformation test specified in 4.4.6 without visual evidence of rupture.

3.1.4.2 Nuts. -

3.1.4.2.1 Tensile strength. - Carbon and alloy steel and corrosion resisting (condition A) nuts shall conform to the applicable strip load requirements specified in tables III and IV when tested in accordance with 4.4.10. The tensile strength of nonferrous nuts shall be determined as specified in 4.4.10. The minimum stripping load shall be that load obtained by multiplying the stress area of the

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mating external thread specified in tables III and IV as applicable, by the minimum tensile strength (in p. s. i.) specified in the applicable materials specification for the grade, class and condition of the metal from which the nut is made.

Table III - Minimum stripping loads of steel nuts, coarse-thread series.

Nominal size or basic major diameter of thread	Threads per inch and designation	Stress area of mating external thread	Minimum breaking or stripping loads for steels [1], [2]					
			Grade 2 69,000	Grade 5			Grade 8 150,000	FS 303 cond. A
				120,000	115,000	105,000		
Inches		Square inches	Pounds	Pounds	Pounds	Pounds	Pounds	Pounds
1/4	20 UNC	0.0318	2,190	3,820	------	-------	4,770	1,750
5/16	18 UNC	.0524	3,620	6,290	------	-------	7,860	2,880
3/8	16 UNC	.0775	5,350	9,300	------	-------	11,620	4,260
7/16	14 UNC	.1063	7,330	12,760	------	-------	15,940	5,850
1/2	13 UNC	.1419	9,790	17,030	------	-------	21,300	7,800
9/16	12 UNC	.182	12,600	21,800	------	-------	27,300	10,010
5/8	11 UNC	.226	15,600	27,100	------	-------	33,900	12,400
3/4	10 UNC	.334	23,000	40,100	------	-------	50,100	18,400
7/8	9 UNC	.462	31,900	------	53,100	-------	69,300	25,400
1	8 UNC	.606	41,800	------	69,700	-------	90,900	33,300
1-1/8	7 UNC	.763	52,600	------	------	80,000	114,000	42,000
1-1/4	7 UNC	.969	66,900	------	------	101,700	145,000	53,300
1-3/8	6 UNC	1.155	79,700	------	------	121,300	173,000	63,500
1-1/2	6 UNC	1.405	96,900	------	------	147,500	211,000	77,300
1-3/4	5 UNC	1.90	131,100	------	------	-------	-------	104,500
2	4-1/2 UNC	2.50	172,500	------	------	-------	-------	137,500
2-1/4	4-1/2 UNC	3.25	224,200	------	------	-------	-------	178,800
2-1/2	4 UNC	4.00	276,000	------	------	-------	-------	220,000
2-3/4	4 UNC	4.93	340,200	------	------	-------	-------	271,200
3	4 UNC	5.97	412,000	------	------	-------	-------	328,000
3-1/4	4 UNC	7.10	490,000	------	------	-------	-------	390,000
3-1/2	4 UNC	8.33	575,000	------	------	-------	-------	458,000
3-3/4	4 UNC	9.66	667,000	------	------	-------	-------	531,000
4	4 UNC	11.08	765,000	------	------	-------	-------	609,000

[1] For slotted and castle nuts take 80 percent of the tabulated loads.
[2] For jam nuts take 46 percent of the tabulated loads.

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Table IV - Minimum stripping loads of steel nuts, fine-thread series.

Nominal size of basic major diameter of thread	Threads per inch and designation	Stress area of mating external thread	Minimum breaking or stripping loads for steels [1], [2]					
			Grade 2	Grade 5			Grade 8	FS 303 cond. A
			69,000	120,000	115,000	105,000	150,000	
Inches		Square inches	Pounds	Pounds	Pounds	Pounds	Pounds	Pounds
1/4	28 UNF	0.0364	2,510	4,370	------	-------	5,460	2,000
5/16	24 UNF	.0580	4,000	6,960	------	-------	8,700	3,190
3/8	24 UNF	.0878	6,060	10,540	------	-------	13,170	4,830
7/16	24 UNF	.1187	8,190	14,240	------	-------	17,800	6,530
1/2	20 UNF	.1599	11,030	19,190	------	-------	24,000	8,790
9/16	18 UNF	.203	14,000	24,400	------	-------	30,400	11,160
5/8	18 UNF	.256	17,700	30,700	------	-------	38,400	14,100
3/4	16 UNF	.373	25,700	44,800	------	-------	56,000	20,500
7/8	14 UNF	.509	35,100	------	58,500	-------	76,400	28,000
1	12 UNF	.663	45,700	------	76,200	-------	99,400	36,500
1-1/8	12 UNF	.856	59,100	------	------	89,900	128,400	47,100
1-1/4	12 UNF	1.073	74,000	------	------	112,700	161,000	59,000
1-3/8	12 UNF	1.315	90,700	------	------	138,100	197,200	72,300
1-1/2	12 UNF	1.581	109,100	------	------	166,000	237,200	87,000

[1]For slotted and castle nuts take 80 percent of the tabulated loads.
[2]For jam nuts take 46 percent of the tabulated loads.

3.1.4.2.2 Hardness. - Carbon and alloy steel nuts, except case hardened nuts, shall conform to table I, as applicable, when tested in accordance with 4.4.2.

3.1.4.2.3 Flattening. - Nuts of ferrous and nonferrous metals shall withstand cold flattening, axially, to one half of the original thickness without visual evidence of cracks, splits, or rupture when tested as specified in 4.4.9. (Case hardened nuts shall not be subjected to this test.)

3.2 Protective coatings. -

3.2.1 Finish. - When bolts, nuts or studs are required to be coated the material shall be cadmium or zinc at the option of the contractor unless a particular material is specified in the contract or order. The coating shall be adherent, smooth and continuous, and shall cover all surfaces including the threads (except the threads of nuts). It shall be free from defects that effect the serviceability or protection afforded the base metal.

3.2.2 Zinc. - The zinc coatings shall conform to type II, class 2 or 3 of Specification QQ-Z-325 as specified (see 6.2).

3.2.2.1 Hot dip zinc coated (galvanized). - Bolts, nuts and studs larger than 1/2 inch size may be coated by hot dipping (galvanized) in lieu of electro-disposition at the contractors option. The coating thickness of the unthreaded portions of bolts, nuts and studs shall conform to the following (see 4.7):

(a) Minimum average thickness shall be 0.002 inch (approx. 1.25 oz. per sq. ft. of surface).
(b) Minimum thickness shall be 0.0015 inch (approx. 0.9 oz. per sq. ft. of surface).

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3.2.2.2 Threads. - When coated by the heat dip process the threads of bolts and studs may be rerolled, but shall not be recut or refinished. Nuts shall be threaded after coating.

3.2.3 Cadmium. - Cadmium coatings shall conform to type II, class B or C of Specification QQ-P-416 as specified (see 6.2).

3.2.4 Salt spray. - When specified, (see 6.2) bolts, nuts and studs shall withstand the salt spray (fog) test specified in Specification QQ-M-151. The zinc coatings shall withstand an exposure of 96 hours, and cadmium plating an exposure of 96 hours without showing evidence of iron rust or white salts except at the root of threads.

3.3 Screw threads. -

3.3.1 Bolts and nuts. - Threads on bolts and nuts shall be of the coarse thread series unified thread form classes 2A or 2B respectively, unless the fine thread series, or 8 thread series or other thread class are specified (see 6.2), the thread form, dimensions, tolerances, and allowances shall be in accordance with Handbook H28.

3.3.2 Studs. - Threads on studs shall be of the unified thread form on the nut end and of the American National form on the tap end. The thread forms, dimensions and tolerances shall be in accordance with Handbook H28. The combinations of threads on studs shall be as follows:

Style	Tap end	Nut end
a	NC-5	UNC-3
b	NC-5	UNC-2

3.3.2.1 Class 5 threads. - The form, dimensions and tolerances of class 5 threads shall be in accordance with Handbook H28.

3.3.3 American National threads. - For an interim period (until July 1958) bolts, nuts, and studs threaded to either the American National or unified thread standards shall be equally acceptable. The thread form, dimensions and tolerances of the American National Standard thread shall be in accordance with Handbook H28. The American National threads corresponding to the unified thread classes are as follows:

American National		Unified
Class 1	or	Class 1A-1B
Class 2	or	Class 2A-2B
Class 3	or	Class 3A-3B
Class 5	or	None

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3.3.4 Thread series. - The number of threads per inch for the unified and American National threads shall be as specified in table V.

Table V - Unified and American coarse and unified and American fine thread series.

Bolt size	Threads per inch		Bolt size	Threads per inch	
	Coarse	Fine		Coarse	Fine
No. 10	24	32	1 inch	8	(14 NF) - (12 UNF)
No. 12	24	28	1-1/8 inches	7	12
1/4 inch	20	28	1-1/4 inches	7	12
5/16 inch	18	24	1-3/8 inches	6	12
3/8 inch	16	24	1-1/2 inches	6	12
7/16 inch	14	20	1-3/4 inches	5	--------
1/2 inch	13	20	2 inches	4-1/2	--------
9/16 inch	12	18	2-1/4 inches	4-1/2	--------
5/8 inch	11	18	2-1/2 inches	4	--------
3/4 inch	10	16	2-3/4 inches	4	--------
7/8 inch	9	14	3 inches	4	--------

3.3.5 Coated threads. - The maximum major pitch and minor diameters may be increased by the amount of the class 2A allowance. Such threads shall be gaged with basic sizes go gages to assure the threads do not exceed the maximum metal limits.

3.3.6 Gaging of threads. - Threads shall be gaged in accordance with the standards and practices specified in Handbook H28 for class "Y" gages.

3.3.6.1 Class 5 fit. - The pitch diameter of class 5 threads shall be measured by means of a cone point snap gage or thread micrometers. The major diameter shall be gaged by means of GO and NOT GO plain ring or snap gages.

3.3.6.2 Class 2A, 2B, and 3A, 3B threads (American National classes 2 and 3). -

3.3.6.2.1 Pitch diameter. - The pitch diameter of classes 2A, 2B and 3A, 3B threads shall be gaged by means of GO and NOT GO thread ring and thread plug gages.

3.3.6.2.2 Major diameter. - The major diameter of class 2A and 3A threads on bolts and studs shall be gaged by means of GO and NOT GO plain ring gages.

3.3.6.2.3 Minor diameter. - The minor diameter of class 2B and 3B threads on nuts shall be gaged by means of GO and NOT GO plain plug gages.

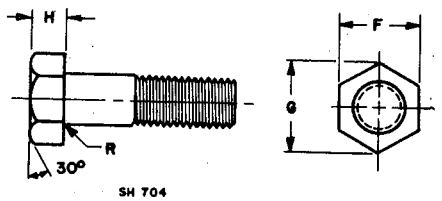
3.4 Dimensions and tolerances of bolts, nuts and studs, other than threads. - The nominal size of bolts, nuts, and studs and nominal length of bolts and studs shall be as specified (see 6.2). Bolts and nuts are normally ordered separately. When ordered together, the class and finish of the nut shall be the same as that of the bolt, unless otherwise specified in the contract or order. When coated bolts and nuts are ordered together, they shall be assembled prior to shipment.

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3.4.1 Bolts.-

3.4.1.1 Limits of size.- Bolts, type I through 5 shall conform to the [illegible] ied in tables VI through XI, and the requirements of 3.4.1.2 through 3.4.1.10.

Table VI - Type I regular unfinished hexagon head bolt.



SH 704

Nominal size or basic major diameter of thread		Body diameter[1]	Width across flats F			Width across corners G		Height H			Radius of fillet R
		Max.	Max.	(basic)	Min.	Max.	Min.	Nom.		Min.	Max.
1/4	0.2500	0.260	7/16	0.4375	0.425	0.505	0.484	11/64	0.188	0.150	0.031
5/16	.3125	.324	1/2	.5000	.484	.577	.552	7/32	.235	.195	.031
3/8	.3750	.388	9/16	.5625	.544	.650	.620	1/4	.268	.226	.031
7/16	.4375	.452	5/8	.6250	.603	.722	.687	19/64	.316	.272	.031
1/2	.5000	.515	3/4	.7500	.725	.866	.826	11/32	.364	.302	.031
5/8	.6250	.642	15/16	.9375	.906	1.083	1.033	27/64	.444	.378	.062
3/4	.7500	.768	1- 1/8	1.1250	1.088	1.299	1.240	1/2	.524	.455	.062
7/8	.8750	.895	1- 5/16	1.3125	1.269	1.516	1.447	37/64	.604	.531	.062
1	1.0000	1.022	1- 1/2	1.5000	1.450	1.732	1.653	43/64	.700	.591	.093
1-1/8	1.1250	1.149	1-11/16	1.6875	1.631	1.949	1.859	3/4	.780	.658	.093
1-1/4	1.2500	1.277	1- 7/8	1.8750	1.812	2.165	2.066	27/32	.876	.749	.093
1-3/8	1.3750	1.404	2- 1/16	2.0625	1.994	2.382	2.273	29/32	.940	.810	.093
1-1/2	1.5000	1.531	2- 1/4	2.2500	2.175	2.598	2.480	1	1.036	.902	.093
1-3/4	1.7500	1.785	2- 5/8	2.6250	2.538	3.031	2.893	1- 5/32	1.196	1.054	.125
2	2.0000	2.039	3	3.0000	2.900	3.464	3.306	1-11/32	1.388	1.175	.125
2-1/4	2.2500	2.305	3- 3/8	3.3750	3.262	3.897	3.719	1- 1/2	1.548	1.327	.188
2-1/2	2.5000	2.559	3- 3/4	3.7500	3.625	4.330	4.133	1-21/32	1.708	1.479	.188
2-3/4	2.7500	2.827	4- 1/8	4.1250	3.988	4.763	4.546	1-13/16	1.869	1.632	.188
3	3.0000	3.081	4- 1/2	4.5000	4.350	5.196	4.959	2	2.060	1.815	.188
3-1/4	3.2500	3.335	4- 7/8	4.8750	4.712	5.629	5.372	2- 3/16	2.251	1.936	.188
3-1/2	3.5000	3.589	5- 1/4	5.2500	5.075	6.062	5.786	2- 5/16	2.380	2.057	.188
3-3/4	3.7500	3.858	5- 5/8	5.6250	5.437	6.495	6.198	2- 1/2	2.572	2.241	.188
4	4.0000	4.111	6	6.0000	5.800	6.928	6.612	2-11/16	2.764	2.424	.188

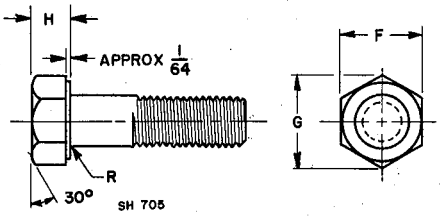
[1]There may be a reasonable swell or fin under the head or die seam on the body not to exceed the nominal body diameter by the following:

0.030 for sizes up to 1/2 inch, 0.050 for sizes 5/8 and 3/4 inch, 0.063 for sizes over 3/4 inch up to 1-1/4 inches,
0.093 for sizes over 1-1/4 inches up to 2 inches, 0.125 for sizes over 2 inches up to 3 inches, and 0.188 for sizes over 3 inches.

NOTES:
1. All dimensions given in inches.
2. Bolt is not finished on any surface.
3. Taper of head (angle between one side and axis) shall not exceed 2 degrees, specified width across flats being the largest dimension.
4. Top of bolt head shall be flat and chamfered. Diameter of top circle shall be maximum width across flats, within a tolerance of minus 15 percent.
5. Bearing surface shall be at right angle to axis of body within a tolerance of 3 degrees for 1-inch size or smaller and 2 degrees for sizes larger than 1 inch. The bearing surface shall be concentric with axis of body within a tolerance of 3 percent of maximum width across flats.
6. Minimum thread length shall be twice the diameter plus 1/4 inch for lengths up to and including 6 inches and twice the diameter plus 1/2 inch for lengths over 6 inches. Bolts too short for the formula thread length shall be threaded as close to the head as practical.
7. Tolerance on bolt length for bolts 6 inches and under in length shall be plus or minus 1/32 inch for sizes 1/4 to 3/8 inch, plus or minus 1/16 inch for sizes 7/16 and 1/2 inch, plus or minus 1/8 inch for sizes 5/8 to 1-1/4 inch and plus or minus 1/4 inch for sizes 1-3/8 to 4 inches. Length tolerance for bolts over 6 inches, in length shall be plus or minus 1/16 inch for sizes 1/4 to 3/8 inch plus or minus 3/32 inch for sizes 7/16 and 1/2 inch, plus or minus 3/16 inch for sizes 5/8 to 1-1/4 inches, and plus or minus 1/4 inch for sizes 1-3/8 to 4 inches.
8. Bolts shall enter ring gage to head; ring gage shall have a maximum thickness of one diameter, a minimum fillet equal to the maximum fillet of the bolt, and an internal diameter equal to the specified maximum body diameter.

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Table VII - Type II regular semifinished hexagon head bolts.



Nominal size or basic major diameter of thread		Body diameter[1]	Width across flats F			Width across corners G		Height H			Radius of fillet R	
		Max.	Max.	(basic)	Min.	Max.	Min.	Nom.	Max.	Min.	Min.	Max.
1/4	0.2500	0.260	7/16	0.4375	0.425	0.505	0.484	5/32	0.163	0.150	0.009	0.031
5/16	.3125	.324	1/2	.5000	.484	.577	.552	13/64	.211	.195	.009	.031
3/8	.3750	.388	9/16	.5625	.544	.650	.620	15/64	.243	.226	.009	.031
7/16	.4375	.452	5/8	.6250	.603	.722	.687	9/32	.291	.272	.009	.031
1/2	.5000	.515	3/4	.7500	.725	.866	.826	5/16	.323	.302	.009	.031
5/8	.6250	.642	15/16	.9375	.906	1.083	1.033	25/64	.403	.378	.021	.062
3/4	.7500	.768	1- 1/8	1.1250	1.088	1.299	1.240	15/32	.483	.455	.021	.062
7/8	.8750	.895	1- 5/16	1.3125	1.269	1.516	1.447	35/64	.563	.531	.031	.062
1	1.0000	1.022	1- 1/2	1.5000	1.450	1.732	1.653	39/64	.627	.591	.062	.093
1-1/8	1.1250	1.149	1-11/16	1.6875	1.631	1.949	1.859	11/16	.718	.658	.062	.093
1-1/4	1.2500	1.277	1- 7/8	1.8750	1.812	2.165	2.066	25/32	.813	.749	.062	.093
1-3/8	1.3750	1.404	2- 1/16	2.0625	1.994	2.382	2.273	27/32	.878	.810	.062	.093
1-1/2	1.5000	1.531	2- 1/4	2.2500	2.175	2.598	2.480	15/16	.974	.902	.062	.093
1-3/4	1.7500	1.785	2- 5/8	2.6250	2.538	3.031	2.893	1- 3/32	1.134	1.054	.078	.125
2	2.0000	2.039	3	3.0000	2.900	3.464	3.306	1- 7/32	1.263	1.175	.078	.125
2-1/4	2.2500	2.305	3- 3/8	3.3750	3.262	3.897	3.719	1- 3/8	1.423	1.327	.125	.188
2-1/2	2.5000	2.559	3- 3/4	3.7500	3.625	4.330	4.133	1-17/32	1.583	1.479	.125	.188
2-3/4	2.7500	2.827	4- 1/8	4.1250	3.988	4.763	4.546	1-11/16	1.744	1.632	.125	.188
3	3.0000	3.081	4- 1/2	4.5000	4.350	5.196	4.959	1- 7/8	1.935	1.815	.125	.188
3-1/4	3.2500	3.335	4- 7/8	4.8750	4.712	5.629	5.372	2	2.064	1.936	.125	.188
3-1/2	3.5000	3.589	5- 1/4	5.2500	5.075	6.062	5.786	2- 1/8	2.193	2.057	.125	.188
3-3/4	3.7500	3.858	5- 5/8	5.6250	5.437	6.495	6.198	2- 5/16	2.385	2.241	.125	.188
4	4.0000	4.111	6	6.0000	5.800	6.928	6.612	2- 1/2	2.576	2.424	.125	.188

[1]There may be a reasonable swell or fin under the head or die seam on the body not to exceed the nominal body diameter by the following:

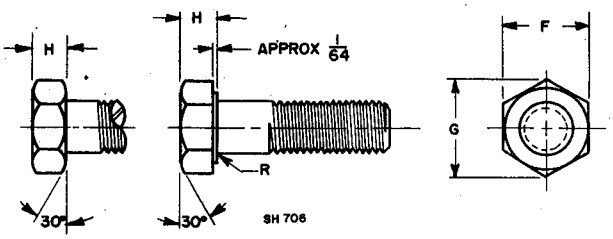
0.030 for sizes up to 1/2 inch, 0.050 for sizes 5/8 and 3/4 inch, 0.063 for sizes over 3/4 inch up to 1-1/4 inches,
0.093 for sizes over 1-1/4 inches up to 2 inches, 0.125 for sizes over 2 inches up to 3 inches, and 0.188 for sizes over 3 inches.

NOTES:

1. All dimensions given in inches.
2. Semifinished bolt is processed to produce a flat bearing surface under head only.
3. Taper of head (angle between one side and axis) shall not exceed 2 degrees, specified width across flats being the largest dimension.
4. Top of head shall be flat and chamfered. Diameter of top circle shall be maximum width across flats within a tolerance of minus 15 percent.
5. Bearing surface shall be flat and washer faced. Diameter of washer face shall be 95 percent of maximum width across flats within a tolerance of plus or minus 5 percent.
6. Bearing surface shall be at right angles to axis of body within a tolerance of 2 degrees for sizes up to and including 1 inch, and within a tolerance of 1 degree for sizes larger than 1 inch. The bearing surface shall be concentric with axis of body within a tolerance of 3 percent of the maximum width across flats.
7. Minimum thread length shall be twice the diameter plus 1/4 inch for lengths up to and including 6 inches; twice the diameter plus 1/2 inch for lengths over 6 inches. The tolerance shall be plus 3/16 inch or 2-1/2 threads, whichever is greater. On products that are too short for minimum thread lengths, the distance from the bearing surface of the head to the first complete thread shall not exceed the length of 2-1/2 threads, as measured with a ring thread gage, for sizes up to and including 1 inch and 3-1/2 threads for sizes larger than 1 inch.
8. Point shall be flat and chamfered or rounded at manufacturer's option, length of point not to exceed length of 1-1/2 threads from extreme point to first full thread.
9. Tolerance on bolt length for bolts 6 inches and under in length shall be plus or minus 1/32 inch for sizes 1/4 to 3/8 inch, plus or minus 1/16 inch for sizes 7/16 and 1/2 inch, plus or minus 1/8 inch for sizes 5/8 to 1-1/4 inches, and plus or minus 1/4 inch for sizes 1-3/8 to 4 inches. Length tolerance for bolts over 6 inches in length shall be plus or minus 1/16 inch for sizes 1/4 to 3/8 inch, plus or minus 3/32 inch for sizes 7/16 and 1/2 inch, plus or minus 3/16 inch for sizes 5/8 to 1-1/4 inches, and plus or minus 1/4 inch for sizes 1-3/8 to 4 inches.
10. Bolts shall enter ring gage to head; ring gage shall have a maximum thickness of one diameter, a minimum fillet equal to the maximum fillet of the bolt, and an internal diameter equal to the specified maximum body diameter.

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Table VIII - Type III regular finished hexagon head bolts.



Nominal size or basic major diameter of thread		Body diameter min. (maximum equal to nominal size	Width across flats F			Width across corners G		Height H			Radius of fillet R	
			Max. (basic)		Min.[1]	Max.	Min.	Nom.	Max.	Min.	Max.	Min.
1/4	0.2500	0.2450	7/16	0.4375	0.428	0.505	0.488	5/32	0.163	0.150	0.009	0.023
5/16	.3125	.3065	1/2	.5000	.489	.577	.557	13/64	.211	.195	.009	.023
3/8	.3750	.3690	9/16	.5625	.551	.650	.628	15/64	.243	.226	.009	.023
7/16	.4375	.4305	5/8	.6250	.612	.722	.698	9/32	.291	.272	.009	.023
1/2	.5000	.4930	3/4	.7500	.736	.866	.840	5/16	.323	.302	.009	.023
9/16	.5625	.5545	13/16	.8125	.798	.938	.910	23/64	.371	.348	.021	.041
5/8	.6250	.6170	15/16	.9375	.922	1.083	1.051	25/64	.403	.378	.021	.041
3/4	.7500	.7410	1- 1/8	1.1250	1.100	1.299	1.254	15/32	.483	.455	.021	.041
7/8	.8750	.8660	1- 5/16	1.3125	1.285	1.516	1.465	35/64	.563	.531	.041	.062
1	1.0000	.9900	1- 1/2	1.5000	1.469	1.732	1.675	39/64	.627	.591	.062	.093
1-1/8	1.1250	1.1140	1-11/16	1.6875	1.631	1.949	1.859	11/16	.718	.658	.062	.093
1-1/4	1.2500	1.2390	1- 7/8	1.8750	1.812	2.165	2.066	25/32	.813	.749	.062	.093
1-3/8	1.3750	1.3630	2- 1/16	2.0625	1.994	2.382	2.273	27/32	.878	.810	.062	.093
1-1/2	1.5000	1.4880	2- 1/4	2.2500	2.175	2.598	2.480	15/16	.974	.902	.062	.093
1-3/4	1.7500	1.7380	2- 5/8	2.6250	2.538	3.031	2.893	1- 3/32	1.134	1.054	.062	.093
2	2.0000	1.9880	3	3.0000	2.900	3.464	3.306	1- 7/32	1.263	1.175	.062	.093
2-1/4	2.2500	2.2380	3- 3/8	3.3750	3.262	3.897	3.719	1- 3/8	1.423	1.327	.062	.093
2-1/2	2.5000	2.4880	3- 3/4	3.7500	3.625	4.330	4.133	1-17/32	1.583	1.479	.062	.093
2-3/4	2.7500	2.7380	4- 1/8	4.1250	3.988	4.763	4.546	1-11/16	1.744	1.632	.062	.093
3	3.0000	2.9880	4- 1/2	4.5000	4.350	5.196	4.959	1- 7/8	1.935	1.815	.062	.093

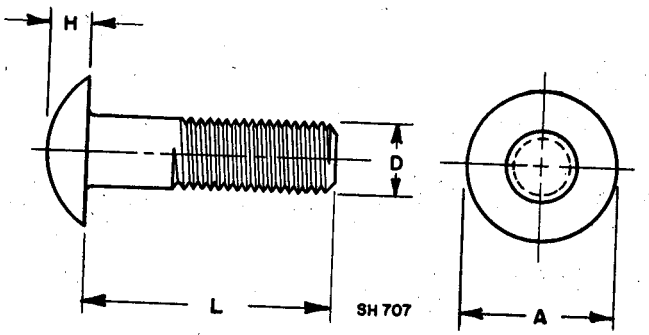
[1]In sizes 1/4 to 1 inch a tolerance of minus 0.050 D may be used when the product in hot made.

NOTES:

1. All dimensions given in inches.
2. Sizes 1/4 through 1 inch are products unified dimensionally with British and Canadian standards.
3. "Finished" in the title refers to the quality of manufacture and the closeness of tolerance and does not indicate that surfaces are completely machined.
4. Taper of head (angle between one side and axis) shall not exceed 2 degrees, specified width across flats being the largest dimension.
5. Top of head shall be flat and chamfered. Diameter of top circle shall be maximum width across flats within a tolerance of minus 15 percent.
6. Bearing surface shall be flat and either washer faced or with chamfered corners. Diameter of bearing surface shall be 95 percent of maximum width across flats within a tolerance of plus or minus 5 percent.
7. Bearing surface shall be at right angles to axis of body within a tolerance of 2 degrees for sizes up to and including 1 inch, and within a tolerance of 1 degree for sizes larger than 1 inch. The bearing surface shall be concentric with axis of body within a tolerance of 3 percent of the maximum width across flats.
8. Minimum thread length shall be twice the diameter plus 1/4 inch for lengths up to and including 6 inches; twice the diameter plus 1/2 inch for lengths over 6 inches. The tolerance shall be plus 3/16 inch or 2-1/2 threads, whichever is greater. On products that are too short for minimum thread lengths, the distance from the bearing surface of the head to the first complete thread shall not exceed the length of 2-1/2 threads, as measured with a ring thread gage, for sizes up to and including 1 inch and 3-1/2 threads for sizes larger than 1 inch.
9. Point shall be flat and chamfered from approximately 1/64 inch below the minor diameter, the length of point to be from 1/2 to 1-1/2 threads.
10. Tolerance on bolt length for sizes up to and including 3/4 inch shall be: minus 1/32 inch for lengths up to and including 1 inch; minus 1/16 inch for lengths over 1 inch to and including 2 inches; minus 3/32 inch for lengths over 2 inches to and including 6 inches; and minus 3/16 inch for lengths over 6 inches. Tolerance for sizes over 3/4 inch shall be: minus 1/16 inch for lengths up to and including 1 inch; minus 1/8 inch for lengths over 1 inch to and including 2 inches; and minus 3/16 inch for lengths over 2 inches.
11. Total runout (eccentricity and angularity) of thread in relation to body. For sizes up to and including 3/4 inch, bolts shall screw at least two threads into a tapped hole counterbored for 1/32 inch diametral clearance over the maximum body diameter to a depth equal to the length of the bolt less one bolt diameter. The starting thread of the tapped hole shall be countersunk to the diameter of the counterbore. The tapped hole shall have a class 2B maximum pitch diameter, and the inspection fixture shall be hardened. For bolts over 3/4 inch, the diametral clearance of the counterbored hole shall be 1/16 inch. A suitable gage is shown in Appendix III.

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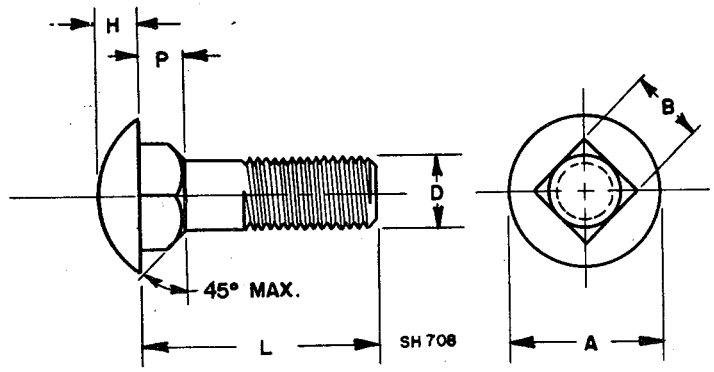
Table IX - Type IV round head, unfinished bolt (button head).



Size		Body diameter	Diameter of head A				Height of head H		
D	Tpi	Max.	Min.	Basic		Max.	Min.		Max.
No. 10	24	0.199	0.438	7/16	0.438	0.469	3/32	0.094	0.114
1/4	20	.260	.563	9/16	.563	.594	1/8	.125	.145
5/16	18	.324	.688	11/16	.688	.719	5/32	.156	.176
3/8	16	.388	.782	13/16	.813	.844	3/16	.188	.208
7/16	14	.452	.907	15/16	.938	.969	7/32	.219	.239
1/2	13	.515	1.032	1- 1/16	1.063	1.094	1/4	.250	.270
5/8	11	.642	1.219	1- 5/16	1.313	1.344	5/16	.313	.344
3/4	10	.768	1.469	1- 9/16	1.563	1.594	3/8	.375	.406
7/8	9	.895	1.719	1-13/16	1.813	1.844	7/16	.438	.469
1	8	1.022	1.969	2- 1/16	2.063	2.094	1/2	.500	.531

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Table X - Type V round head square neck unfinished bolt (square neck carriage bolt).



Size		Body diameter	Diameter of head A			Height of head H		Depth of square P		Width of square B	
D	T.p.i.	Max.	Min.	Basic	Max.	Min.	Max.	Min.	Max.	Min.	Max.
No. 10	24	0.199	0.438	7/16 0.438	0.469	3/32 0.094	0.114	0.094	0.125	0.185	0.199
1/4	20	.260	.563	9/16 .563	.594	1/8 .125	.145	.125	.156	.245	.260
5/16	18	.324	.688	11/16 .688	.719	5/32 .156	.176	.156	.187	.307	.324
3/8	16	.388	.782	13/16 .813	.844	3/16 .188	.208	.188	.219	.368	.388
7/16	14	.452	.907	15/16 .938	.969	7/32 .219	.239	.219	.250	.431	.452
1/2	13	.515	1.032	1- 1/16 1.063	1.094	1/4 .250	.270	.250	.281	.492	.515
5/8	11	.642	1.219	1- 5/16 1.313	1.344	5/16 .313	.344	.313	.344	.616	.642
3/4	10	.768	1.469	1- 9/16 1.563	1.594	3/8 .375	.406	.375	.406	.741	.768
7/8	9	.895	1.719	1-13/16 1.813	1.844	7/16 .438	.469	.438	.469	.865	.895
1	8	1.022	1.969	2- 1/16 2.063	2.094	1/2 .500	.531	.500	.531	.990	1.022

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Table XI - Tolerance on bolt and stud lengths.

Bolt length (inches)	Tolerance for nominal size of bolt (inches)			
	1/4 to 3/8	7/16 and 1/2	5/8 to 1-1/4	Over 1-1/4
Up to 6, incl.	±1/32	±1/16	±1/8	±1/4
Over 6	±1/16	±3/32	±3/16	±1/4

3.4.1.2 Body diameter. - The maximum and minimum limits of the body diameters of bolts shall be as follows:

(a) Types I and II. - The minimum body diameter shall be not less than the nominal pitch diameter of the thread. The maximum body diameter shall be as specified in tables VI and VII.

(b) Type III. - The minimum and maximum body diameters shall be as specified in table VIII.

(c) Types IV and V. - The maximum body diameter shall be as specified in tables IX and X. When specified, (see 6.2) an undersize body having a diameter approximating the pitch diameter of the thread shall be furnished.

3.4.1.3 Length of bolts. - The length of types I, II and III bolts is the distance from the bearing surface of the head to the extreme point. The length of types IV and V bolts is the distance from the largest diameter of the head to the end of the bolt.

3.4.1.3.1 Length tolerance. - Tolerances on the length of bolts shall be as specified in table XI.

3.4.1.4 Point of bolt. - The points of types I, II and III bolts shall be as specified in tables VI, VII and VIII, respectively. For types IV and V bolts, the points may be of sheared end with the first thread incomplete, or of round end, or chamfered 45 degrees to the minor diameter within the limits specified; the points of types IV and V bolts are at the option of the contractor.

3.4.1.5 Fillets. - The radius of fillet for types I, II and III bolts shall be as specified in tables VI, VII and VIII. For types IV and V bolts the maximum radius of fillet shall be 1/32 inch for sizes 1/2 inch and smaller, and 1/16 inch for sizes over 1/2 inch. For bolts made of corrosion resisting steel and nonferrous metals the radius of fillet may be increased somewhat to conform with good manufacturing practice.

3.4.1.6 Length of thread. - The length of thread is the distance from the extreme end of the bolt to the last complete thread. The minimum length of thread for bolts shall be as follows:

(a) For bolts up to and including 6 inches in length the minimum thread length shall be two diameters (nominal size) plus 1/4 inch.

(b) For bolts over 6 inches in length the minimum thread length shall be two diameters (nominal size) plus 1/2 inch.

Bolts too short for the formula thread length specified in (a) and (b) shall be threaded as close to the head or neck as practicable.

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3.4.1.7 Bolt heads. - For types I, II, and III bolts, the bolt head shall conform to the requirements of tables VI, VII and VIII. The height of the bolt head (H) is the overall distance from the top to the bearing surface, and shall include the washer face when provided. The heads of types IV and V bolts are not machined or trimmed, hence the circumference may be somewhat irregular and the edge may be flat or rounded.

3.4.1.8 Washer face. - When provided, the washer face shall be a circular boss machined or otherwise produced on the bolt head to relieve the corners.

3.4.1.9 Concentricity and angularity. - The concentricity and angularity of the bolt head with the longitudinal axis of the bolt, for types I, II, and III bolts shall conform to the requirements of tables VI, VII and VIII.

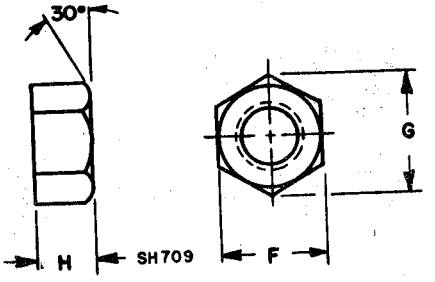
3.4.1.10 Diameter of shank. - Bolts, type I, II, and III shall enter a ring gage to the head. The ring gage shall have a maximum thickness of one nominal diameter of the bolt, a minimum fillet equal to the specified maximum fillet of the bolt, and an internal diameter equal to the specified body diameter.

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3.4.2 Nuts. -

3.4.2.1 Limits of size. - Nuts shall conform to the dimensions and tolerances and other requirements specified in tables XII to XVI inclusive.

Table XII - Type I regular hexagon nuts (unfinished) 1/4 to 5/8 inch sizes not recommended for new designs.[1]



Nominal size or basic major diameter of thread		Width across flats F			Width across corners G		Thickness regular nuts H		
		Max. (basic)		Min.	Max.	Min.	Nom.	Max.	Min.
1/4	0.2500	7/16	0.4375	0.425	0.505	0.484	7/32	0.235	0.203
5/16	.3125	9/16	.5625	.547	.650	.624	17/64	.283	.249
3/8	.3750	5/8	.6250	.606	.722	.691	21/64	.346	.310
7/16	.4375	3/4	.7500	.728	.866	.830	3/8	.394	.356
1/2	.5000	13/16	.8125	.788	.938	.898	7/16	.458	.418
9/16	.5625	7/8	.8750	.847	1.010	.966	1/2	.521	.479
5/8	.6250	1	1.0000	.969	1.155	1.104	35/64	.569	.525
3/4	.7500	1- 1/8	1.1250	1.088	1.299	1.240	21/32	.680	.632
7/8	.8750	1- 5/16	1.3125	1.269	1.516	1.447	49/64	.792	.740
1	1.0000	1- 1/2	1.5000	1.450	1.732	1.653	7/8	.903	.847
1-1/8	1.1250	1-11/16	1.6875	1.631	1.949	1.859	1	1.030	.970
1-1/4	1.2500	1- 7/8	1.8750	1.812	2.165	2.066	1- 3/32	1.126	1.062
1-3/8	1.3750	2- 1/16	2.0625	1.994	2.382	2.273	1-13/64	1.237	1.169
1-1/2	1.5000	2- 1/4	2.2500	2.175	2.598	2.480	1- 5/16	1.348	1.276

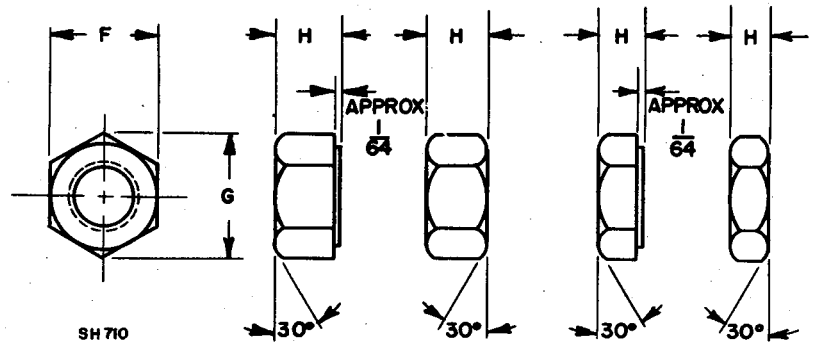
[1]The dimensions of 1/4 to 5/8 inch size nuts are published for interim use during the change-over to the new standard table XIV until such time as complete conversion can be made.

NOTES:

1. All dimensions given in inches.
2. Nuts are not finished on any surface but are threaded.
3. Taper of the sides of nuts (angle between one side and the axis) shall not exceed 2 degrees, the specified width across flats being the largest dimension.
4. Tops of nuts shall be flat and chamfered or (except in jam nuts) washer crowned. Diameter of top circle shall be the maximum width across flats within a tolerance of minus 15 percent.
5. Bearing surface shall be at right angles to the axis of the threaded hole within a tolerance of 3 degrees for 1-inch nuts or smaller and 2 degrees for nuts larger than 1 inch; therefore, the maximum total runout of bearing face would equal the tangent of specified angle times the distance across flats.

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Table XIII - Types II and IV, regular semifinished hexagon and hexagon-jam nuts 1/4 to 5/8 inch sizes not recommended for new designs.[1]



Nominal size or basic major diameter of thread		Width across flats F			Width across corners G		Thickness regular nuts H			Thickness regular jam nuts H		
		Max. (basic)		Min.	Max.	Min.	Nom.	Max.	Min.	Nom.	Max.	Min.
1/4	0.2500	7/16	0.4375	0.425	0.505	0.485	13/64	0.219	0.187	9/64	0.157	0.125
5/16	.3125	9/16	.5625	.547	.650	.624	1/4	.267	.233	11/64	.189	.155
3/8	.3750	5/8	.6250	.606	.722	.691	5/16	.330	.294	13/64	.221	.185
7/16	.4375	3/4	.7500	.728	.866	.830	23/64	.378	.340	15/64	.253	.215
1/2	.5000	13/16	.8125	.788	.938	.898	27/64	.442	.402	19/64	.317	.277
9/16	.5625	7/8	.8750	.847	1.010	.966	31/64	.505	.463	21/64	.349	.307
5/8	.6250	1	1.0000	.969	1.155	1.104	17/32	.553	.509	23/64	.381	.337

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Table XIII - Types II and IV, regular semifinished hexagon and hexagon-jam nuts 1/4 to 5/8 inch sizes not recommended for new designs[1] (cont'd).

Nominal size or basic major diameter of thread		Width across flats F			Width across corners G		Thickness regular nuts H			Thickness regular jam nuts H		
		Max. (basic)		Min.	Max.	Min.	Nom.	Max.	Min.	Nom.	Max.	Min.
Below this line, dimensions are in accord with table XIV (finished hexagon and hexagon jam nuts)												
3/4	0.7500	1- 1/8	1.1250	1.088	1.299	1.240	41/64	0.665	0.617	27/64	0.446	0.398
7/8	.8750	1- 5/16	1.3125	1.269	1.516	1.447	3/4	.776	.724	31/64	.510	.458
1	1.0000	1- 1/2	1.5000	1.450	1.732	1.653	55/64	.887	.831	35/64	.575	.519
1-1/8	1.1250	1-11/16	1.6875	1.631	1.949	1.859	31/32	.999	.939	39/64	.639	.579
1-1/4	1.2500	1- 7/8	1.8750	1.812	2.165	2.066	1-1/16	1.094	1.030	23/32	.751	.687
1-3/8	1.3750	2- 1/16	2.0625	1.994	2.382	2.273	1-11/64	1.206	1.138	25/32	.815	.747
1-1/2	1.5000	2- 1/4	2.2500	2.175	2.598	2.480	1- 9/32	1.317	1.245	27/32	.880	.808
1-3/4	1.7500	2- 5/8	2.6250	2.538	3.031	2.893	1- 1/2	1.540	1.460	31/32	1.009	.929
2	2.0000	3	3.0000	2.900	3.464	3.306	1-23/32	1.763	1.675	1- 3/32	1.138	1.050
2-1/4	2.2500	3- 3/8	3.3750	3.262	3.897	3.719	1-59/64	1.970	1.874	1-13/64	1.251	1.155
2-1/2	2.5000	3- 3/4	3.7500	3.625	4.330	4.133	2- 9/64	2.193	2.089	1-29/64	1.505	1.401
2-3/4	2.7500	4- 1/8	4.1250	3.988	4.763	4.546	2-23/64	2.415	2.303	1-37/64	1.634	1.522
3	3.0000	4- 1/2	4.5000	4.350	5.196	4.959	2-37/64	2.638	2.518	1-45/64	1.763	1.643

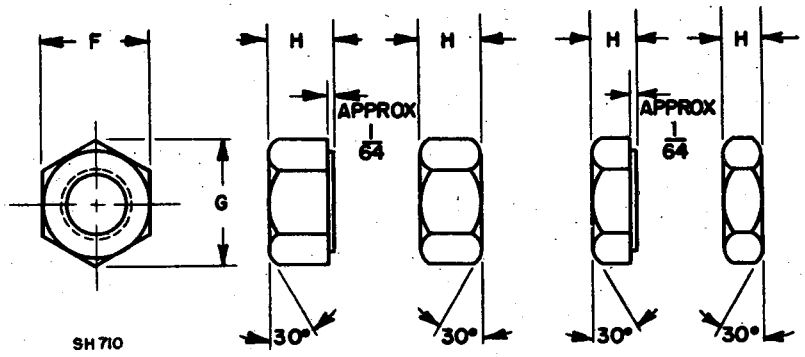
[1]This table is published for interim use during the change-over to the new standard table XIV until such time as complete conversion can be made.

NOTES:

1. All dimensions given in inches.
2. Semifinished nuts are finished on bearing surface and threaded.
3. Taper of the sides of nuts (angle between one side and the axis) shall not exceed 2 degrees, the specified width across flats being the largest dimension.
4. Tops of nuts shall be flat and chamfered. Diameter of top circle shall be the maximum width across flats within a tolerance of minus 15 percent.
5. Bearing surface shall be washer faced or with chamfered corners. Diameter of washer face and the diameter of circle of bearing surface of double chamfered nuts shall be the maximum width across flats within a tolerance of minus 5 percent.
6. Bearing surface shall be at right angles to the axis of the threaded hole within a tolerance of 2 degrees for 5/8 inch nuts or smaller and 1 degree for nuts larger than 5/8 inch; therefore, the maximum total runout of bearing face would equal the tangent of specified angle times the distance across flats.

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Table XIV - Types III and V finished-hexagon and hexagon-jam nuts.



Nominal size or basic major diameter of thread		Width across flats F			Width across corners G		Thickness nuts H			Thickness jam nuts H		
		Max. (basic)		Min.	Max.	Min.	Nom.	Max.	Min.	Nom.	Max.	Min.
1/4	0.2500	7/16	0.4375	0.428	0.505	0.488	7/32	0.226	0.212	5/32	0.163	0.150
5/16	.3125	1/2	.5000	.489	.577	.557	17/64	.273	.258	3/16	.195	.180
3/8	.3750	9/16	.5625	.551	.650	.628	21/64	.337	.320	7/32	.227	.210
7/16	.4375	11/16	.6875	.675	.794	.768	3/8	.385	.365	1/4	.260	.240
1/2	.5000	3/4	.7500	.736	.866	.840	7/16	.448	.427	5/16	.323	.302
9/16	.5625	7/8	.8750	.861	1.010	.982	31/64	.496	.473	6/16	.324	.301
5/8	.6250	15/16	.9375	.922	1.083	1.051	35/64	.559	.535	3/8	.387	.363
3/4	.7500	1- 1/8	1.1250	1.088	1.299	1.240	41/64	.665	.617	27/64	.446	.398
7/8	.8750	1- 5/16	1.3125	1.269	1.516	1.447	3/4	.776	.724	31/64	.510	.458
1	1.0000	1- 1/2	1.5000	1.450	1.732	1.653	55/64	.887	.831	35/64	.575	.519
1-1/8	1.1250	1-11/16	1.6875	1.631	1.949	1.859	31/32	.999	.939	39/64	.639	.579
1-1/4	1.2500	1- 7/8	1.8750	1.812	2.165	2.066	1- 1/16	1.094	1.030	23/32	.751	.687
1-3/8	1.3750	2- 1/16	2.0625	1.994	2.382	2.273	1-11/64	1.206	1.138	25/32	.815	.747

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Table XIV - Types III and V finished hexagon and hexagon-jam nuts (cont'd.).

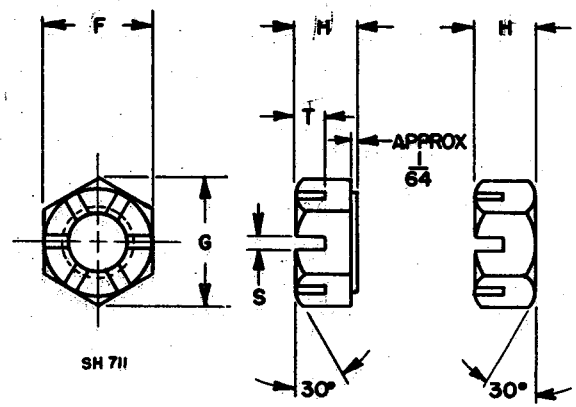
Nominal size or basic major diameter of thread		Width across flats F			Width across corners G		Thickness nuts H			Thickness jam nuts H		
		Max.	(basic)	Min.	Max.	Min.	Nom.	Max.	Min.	Nom.	Max.	Min.
1-1/2	1.5000	2- 1/4	2.2500	2.175	2.598	2.480	1- 9/32	1.317	1.245	27/32	0.880	0.808
1-3/4	1.7500	2- 5/8	2.6250	2.538	3.031	2.893	1- 1/2	1.540	1.460	31/32	1.009	.929
2	2.0000	3	3.0000	2.900	3.464	3.306	1-23/32	1.763	1.675	1- 3/32	1.138	1.050
2-1/4	2.2500	3- 3/8	3.3750	3.262	3.897	3.719	1-59/64	1.970	1.874	1-13/64	1.251	1.155
2-1/2	2.5000	3- 3/4	3.7500	3.625	4.330	4.133	2- 9/64	2.193	2.089	1-29/64	1.505	1.401
2-3/4	2.7500	4- 1/8	4.1250	3.988	4.763	4.546	2-23/64	2.415	2.303	1-37/64	1.634	1.522
3	3.0000	4- 1/2	4.5000	4.350	5.196	4.959	2-37/64	2.638	2.518	1-45/64	1.763	1.643

NOTES:

1. All dimensions given in inches.
2. Sizes 1/4 through 1 inch are products unified dimensionally with British and Canadian standards.
3. "Finished" in the title refers to the quality of manufacture and the closeness of tolerance and does not indicate that surfaces are completely machined.
4. Taper of the sides of nuts (angle between one side and the axis) shall not exceed 2 degrees, the specified width across flats being the largest dimension.
5. Tops of nuts shall be flat and chamfered. Diameter of top circle shall be the maximum width across flats within a tolerance of minus 15 percent for washer faced nuts and within a tolerance of minus 5 percent for double chamfered nuts.
6. Bearing surface shall be washer faced or with chamfered corners. Diameter of circle of bearing surface shall be maximum width across flats within a tolerance of minus 5 percent. Tapped hole shall be countersunk 1/64 inch over the major diameter of thread for nuts up to and including 1/2 inch and 1/32 inch over the major diameter of thread for nuts over 1/2 inch size.
7. Bearing surface shall be at right angles to the axis of the threaded hole within a tolerance of 2 degrees for 5/8 inch nuts or smaller and 1 degree for nuts larger than 5/8 inch; therefore, the maximum total runout of bearing face would equal the tangent of specified angle times the distance across flats.
8. Tolerance on width across flats may be increased 0.015 inch for hot formed nuts 5/8 inch and smaller.

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Table XV - Type VI regular semifinished hexagon slotted nuts 1/4 to 5/8 inch sizes not recommended for new designs.[1]



Nominal size or basic major diameter of thread		Width across flats F			Width across corners G		Thickness H			Slot	
		Max. (basic)		Min.	Max.	Min.	Nom.	Max.	Min.	Width S	Depth T
1/4	0.2500	7/16	0.4375	0.425	0.505	0.485	13/64	0.219	0.187	0.078	0.094
5/16	.3125	9/16	.5625	.547	.650	.624	1/4	.267	.233	.094	.094
3/8	.3750	5/8	.6250	.606	.722	.691	5/16	.330	.294	.125	.125
7/16	.4375	3/4	.7500	.728	.866	.830	23/64	.378	.340	.125	.156
1/2	.5000	13/16	.8125	.788	.938	.898	27/64	.442	.402	.156	.156
9/16	.5625	7/8	.8750	.847	1.010	.966	31/64	.505	.463	.156	.188
5/8	.6250	1	1.0000	.969	1.155	1.104	17/32	.553	.509	.188	.219

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Table XV - Type VI regular semifinished hexagon slotted nuts 1/4 to 5/8 inch sizes not recommended for new designs[1] (cont'd.).

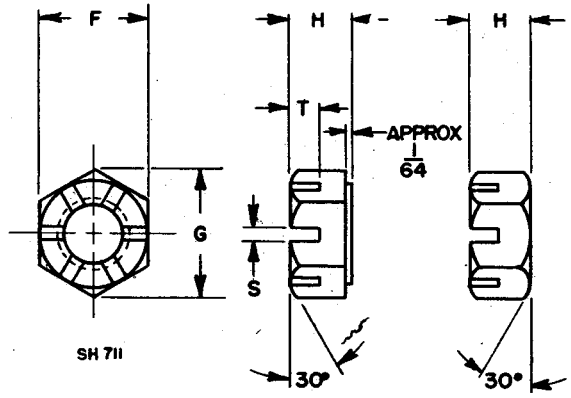
Nominal size or basic major diameter of thread		Width across flats F		Width across corners G		Thickness H			Slot	
		Max. (basic)	Min.	Max.	Min.	Nom.	Max.	Min.	Width S	Depth T
Below this line, dimensions are in accord with table XVI (finished hexagon slotted nut)										
3/4	0.7500	1- 1/8	1.1250	1.088	1.299	1.240	41/64	0.665	0.617	0.188	0.250
7/8	.8750	1- 5/16	1.3125	1.269	1.516	1.447	3/4	.776	.724	.188	.250
1	1.0000	1- 1/2	1.5000	1.450	1.732	1.653	55/64	.887	.831	.250	.281
1-1/8	1.1250	1-11/16	1.6875	1.631	1.949	1.859	31/32	.999	.939	.250	.344
1-1/4	1.2500	1- 7/8	1.8750	1.812	2.165	2.066	1- 1/16	1.094	1.030	.312	.375
1-3/8	1.3750	2- 1/16	2.0625	1.994	2.382	2.273	1-11/64	1.206	1.138	.312	.375
1-1/2	1.5000	2- 1/4	2.2500	2.175	2.598	2.480	1- 9/32	1.317	1.245	.375	.438
1-3/4	1.7500	2- 5/8	2.6250	2.538	3.031	2.893	1- 1/2	1.540	1.460	.438	.500
2	2.0000	3	3.0000	2.900	3.464	3.306	1-23/32	1.763	1.675	.438	.562
2-1/4	2.2500	3- 3/8	3.3750	3.262	3.897	3.719	1-59/64	1.970	1.874	.438	.562
2-1/2	2.5000	3- 3/4	3.7500	3.625	4.330	4.133	2- 9/64	2.193	2.089	.562	.688
2-3/4	2.7500	4- 1/8	4.1250	3.988	4.763	4.546	2-23/64	2.415	2.303	.562	.688
3	3.0000	4- 1/2	4.5000	4.350	5.196	4.959	2-37/64	2.638	2.518	.625	.750

[1]This table is published for interim use during the change-over to the new standard table XVI until such time as complete conversion can be made.

NOTES:
1. All dimensions given in inches.
2. Semifinished nuts are finished on bearing surface and threaded.
3. Taper of the sides of nuts (angle between one side and the axis) shall not exceed 2 degrees, the specified width across flats being the largest dimension.
4. Tops of nuts shall be flat and chamfered. Diameter of top circle shall be the maximum width across flats within a tolerance of minus 15 percent.
5. Bearing surface shall be washer faced or with chamfered corners. Diameter of washer face shall be the maximum width across flats within a tolerance of minus 5 percent.
6. Bearing surface shall be at right angles to the axis of the threaded hole within a tolerance of 2 degrees for 5/8 inch nuts or smaller and 1 degree for nuts larger than 5/8 inch; therefore, the maximum total runout of bearing face would equal the tangent of specified angle times the distance across flats.
7. Slots may have square or round bottoms at option of manufacturer.

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Table XVI - Type VII finished hexagon slotted nuts.



Nominal size or basic major diameter of thread		Width across flats[1] F		Width across corners G		Thickness H			Slot	
		Max. (basic)	Min.	Max.	Min.	Nom.	Max.	Min.	Width S	Depth T
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch
1/4	0.2500	7/16	0.4375	0.428	0.505	0.488	7/32	0.226	0.212	0.078	0.094
5/16	.3125	1/2	.5000	.489	.577	.557	17/64	.273	.258	.094	.094
3/8	.3750	9/16	.5625	.551	.650	.628	21/64	.337	.320	.125	.125
7/16	.4375	11/16	.6875	.675	.794	.768	3/8	.385	.365	.125	.156
1/2	.5000	3/4	.7500	.736	.866	.840	7/16	.448	.427	.156	.156
9/16	.5625	7/8	.8750	.861	1.010	.982	31/64	.496	.473	.156	.188
5/8	.6250	15/16	.9375	.922	1.083	1.051	35/64	.559	.535	.188	.219
3/4	.7500	1- 1/8	1.1250	1.088	1.299	1.240	41/64	.665	.617	.188	.250
7/8	.8750	1- 5/16	1.3125	1.269	1.516	1.447	3/4	.776	.724	.188	.250
1	1.0000	1- 1/2	1.5000	1.450	1.732	1.653	55/64	.887	.831	.250	.281
1-1/8	1.1250	1-11/16	1.6875	1.631	1.949	1.859	31/32	.999	.939	.250	.344
1-1/4	1.2500	1- 7/8	1.8750	1.812	2.165	2.066	1- 1/16	1.094	1.030	.312	.375
1-3/8	1.3750	2- 1/16	2.0625	1.994	2.382	2.273	1-11/64	1.206	1.138	.312	.375
1-1/2	1.5000	2- 1/4	2.2500	2.175	2.598	2.480	1- 9/32	1.317	1.245	.375	.438
1-3/4	1.7500	2- 5/8	2.6250	2.538	3.031	2.893	1- 1/2	1.540	1.460	.438	.500

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Table XVI - Type VII finished hexagon slotted nuts (cont'd.).

Nominal size or basic major diameter of thread	Width across flats[1] F			Width across corners G		Thickness H			Slot	
	Max. (basic)		Min.	Max.	Min.	Nom.	Max.	Min.	Width S	Depth T
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch
2	2.0000	3	3.0000	2.900	3.464	3.306	1-23/32	1.763	1.675	0.438	0.562
2-1/4	2.2500	3- 3/8	3.3750	3.262	3.897	3.719	1-59/64	1.970	1.874	.438	.562
2-1/2	2.5000	3- 3/4	3.7500	3.625	4.330	4.133	2- 9/64	2.193	2.089	.562	.688
2-3/4	2.7500	4- 1/8	4.1250	3.988	4.763	4.546	2-23/64	2.415	2.303	.562	.688
3	3.0000	4- 1/2	4.5000	4.350	5.196	4.959	2-37/64	2.638	2.518	.625	.750

[1]Tolerance on width across flats may be increased 0.015 inch for hot formed nuts 5/8 inch or smaller.

NOTES:

1. Finish.- "Finished" in the title refers to the quality of the manufacture and closeness of tolerance and does not indicate that surfaces are completely machined.
2. Top of nut.- The top shall be flat and chamfered. The diameter of the top circle shall be the maximum width across flats within a tolerance of minus 15 percent for washer faced nuts and within a tolerance of minus 5 percent for double chamfered nuts.
3. Bearing surface.- The bearing surface shall be either washer faced or have chamfered corners. The diameter of the washer face and the diameter of the circle of the bearing surface of double chamfered nuts shall be the maximum width across flats within a tolerance of minus 5 percent. The tapped hole shall be countersunk 1/64 inch over the major diameter of the thread for nuts up to and including 1/2 inch and 1/32 inch over the major diameter of the thread for nuts over 1/2-inch size.
4. The bearing surface shall be at a right angle to the axis of the threaded hole within a tolerance of 2 degrees for nuts 5/8 inch or smaller and 1 degree for nuts larger than 5/8 inch. Accordingly, the maximum total runout of the bearing face would equal the tangent of the specified angle times the width across flats.
5. Side taper.- The taper of the side (angle between one side and the axis) shall not exceed 2 degrees, the specified width across flats being the largest dimension.
6. Slots.- The slots may have square or round bottoms at the option of the manufacturer.
7. Sizes 1/4 through 1 inch are products unified dimensionally with British and Canadian standards.

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3.4.2.1.1 Hot-dip zinc coated. - When nuts are coated by the hot dip process (galvanized) an additional plus tolerance of 0.010 inch, over and above the limits specified in tables II to XVI inclusive shall be permitted on all external dimensions.

3.4.2.2 Thickness of nut. - The thickness of nut is the overall distance from the top to the bearing surface and shall include the thickness of the washer face where provided.

3.4.2.3 Washer face. - The washer face is a circular boss turned or otherwise produced on the bearing surface of the nut to relieve the corners.

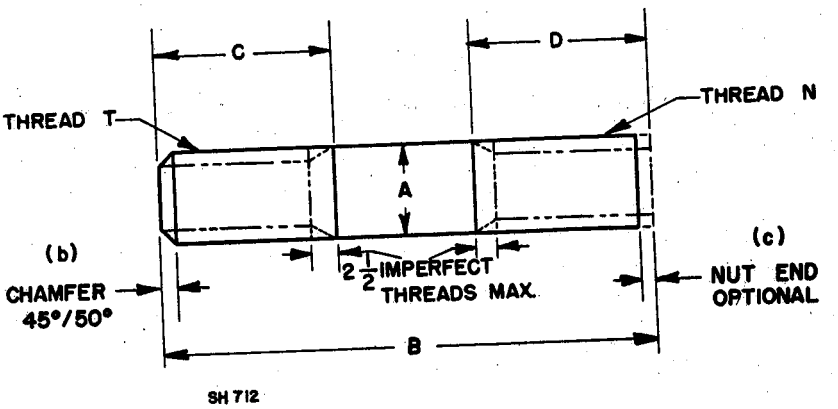
3.4.2.4 Unified dimensions. - The dimensions of nuts in sizes 1/4 to 5/8 inch inclusive specified in tables XII, XIII and XV are published for interim use until industry can convert production to the unified standards indicated in tables XIV and XVI. Nuts produced to either the interim or unified standards are equally acceptable when either is ordered (see 6.6).

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3.4.3 Studs. -

3.4.3.1 Limits of size dimensions and tolerances for studs types I and II, styles a and b shall be in accordance with tables XI, XVII and XVIII.

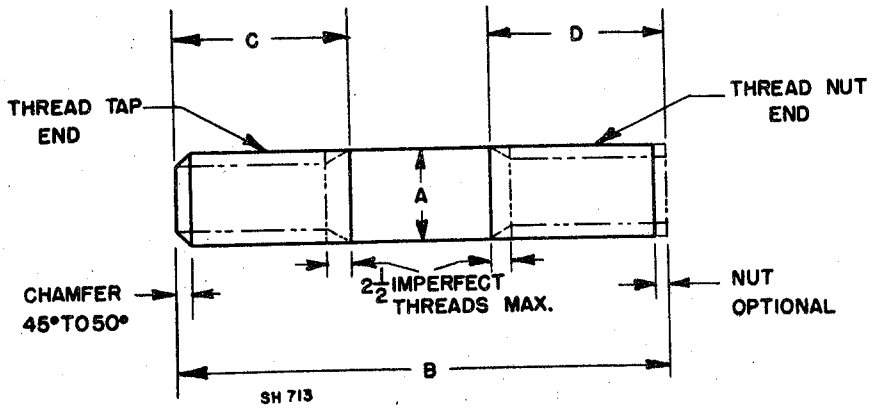
Table XVII - Body diameter - studs types I and II, styles a and b.



Size	Body diameter (inches)							
	Full body				Reduced body			
	Type I, style a		Type I, style b		Type II, style a		Type II, style b	
	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.
1/4	0.270	0.241	0.270	0.240	0.250	0.214	0.248	0.212
5/16	.332	.303	.332	.302	.312	.273	.311	.271
3/8	.395	.365	.395	.364	.375	.331	.373	.328
7/16	.457	.427	.457	.425	.437	.387	.436	.385
1/2	.520	.489	.520	.487	.500	.446	.498	.443
9/16	.582	.551	.582	.549	.562	.504	.560	.501
5/8	.645	.612	.645	.611	.625	.561	.623	.558
3/4	.770	.737	.770	.735	.75[illegible]	.680	.748	.677
7/8	.895	.861	.895	.859	.875	.798	.873	.794
1	1.020	.985	1.020	.983	1.000	.913	.998	.910
1-1/8	1.145	1.108	1.145	1.106	1.12[illegible]	1.026	1.122	1.022
1-1/4	1.270	1.233	1.270	1.231	1.250	1.151	1.247	1.147
1-3/8	1.295	1.356	1.295	1.354	1.375	1.260	1.372	1.256
1-1/2	1.520	1.481	1.520	1.479	1.500	1.385	1.497	1.381

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Table XVIII - Length of thread - studs types I and II, styles a and b.



Nominal size			Minimum thread length - (inches)		
	T. p. i.		Nut end (D)		
Inch	Coarse thread series	Fine thread series	Up to 6 inches long (B)	Over 6 inches long (B)	Tap end (C)
1/4	20	28	0.750	1.000	0.375
5/16	18	24	.875	1.125	.469
3/8	16	24	1.000	1.250	.562
7/16	14	20	1.125	1.375	.656
1/2	13	20	1.250	1.500	.750
9/16	12	18	1.375	1.625	.844
5/8	11	18	1.500	1.750	.937
3/4	10	16	1.750	2.000	1.125
7/8	9	14	2.000	2.250	1.312
1	8	12	2.250	2.500	1.500
1-1/8	7	12	2.500	2.750	1.687
1-1/4	7	12	2.750	3.000	1.875
1-1/2	6	12	3.250	3.500	2.500

3.4.3.2 Body diameters. - The body diameters of types I and II studs are defined as follows:

(a) Full body. - The body diameter (A) is not less than the minimum major diameter of the thread of the nut end nor more than the nominal (basic) size plus 0.020 inch.

(b) Reduced body. - The body diameter (A) is not less than the minimum pitch diameter nor more than the maximum major diameter of the nut end thread.

The body diameter limits for styles a and b studs shall be as specified in table XVII.

3.4.3.3 Length of thread. - The length of thread shall be measured from the extreme ends of the studs and is the distance from the respective ends to the thread runout.

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3.4.3.3.1 Styles a and b.- The maximum and minimum length of thread of the tap end and nut end for studs types I and II, styles a and b shall be as follows:

End of stud	Length of stud	Minimum length of thread	Maximum length of thread
Nut end	6 inches and less	Two times the nominal diameter plus 1/4 inch	Minimum length plus 3/16 inch or 2-1/2 threads whichever is greater
Nut end	Over 6 inches	Two times the nominal diameter plus 1/2 inch	Minimum length plus 3/16 inch or 2-1/2 threads whichever is greater
Tap end	All lengths	1-1/2 times the nominal diameter	Minimum length plus 3/16 inch or 2-1/2 threads, whichever is greater

3.4.3.3.2 When studs are too short to be threaded as specified, the threading shall be of equal length from each end to the center and the ends shall be rounded or chamfered.

3.4.3.4 Rolled threads.- Rolled threads shall be produced on finished blanks by a single rolling from each end. Flow lines of the threads shall be continuous and shall follow the general thread contour.

3.4.3.5 Cut threads.- Machined or die cut threads conforming to this specification are acceptable.

3.4.3.6 Thread taper.- Taper variations of the thread shall fall within the applicable pitch diameter tolerances.

3.4.3.7 Straightness.- When rolled on a surface plate, the clearance as measured by a feeler gage shall not exceed the product:

$$\frac{B}{5D} \times T$$

Where:
B is the nominal length,
D is the nominal diameter and
T is the pitch diameter tolerance of the nut end thread.

3.4.3.8 Ends of studs.- The tap ends of studs shall be chamfered 45 to 50 degrees from a point 1/64 to 1/32 inch below the root diameter of the thread. The shape of the nut end is optional, except as specified in 3.4.3.3.3, but shall fall within the same dimensional envelope as the tap end.

3.4.3.9 Length tolerances.- The length tolerance for studs types I and II, styles a and b shall be as specified in table XI.

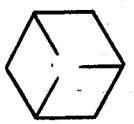
3.4.10 Finish.- Unless otherwise specified in the contract or order, studs shall be furnished unfinished.

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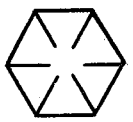
3.5 Identification marking. -

3.5.1 Bolts and studs. - Carbon and alloy steel bolts and studs shall be marked with the grade identification symbols specified below. The symbols shall be applied to the heads of bolts and nut end of studs.

Grades 1 and 2 - No grade identification marking is required.
Grade 5 - Three radial dashes 120 degrees apart.
Grade 8 - Six radial dashes 60 degrees apart.



GRADE 5.



GRADE 8.

SH 714

3.5.2 Nuts. - Carbon and alloy steel nuts shall be marked with the grade identification symbols given below. The markings shall be applied on the top face (not the washer face) of the nut. Slotted nuts are excluded from this requirement.

Grade 1 - No identification marking required.
Grade 2 - No grade marking is required.
Grade 5 - Three raised dashes spaced 120 degrees apart.
Grade 8 - Six raised dashes spaced 60 degrees apart.

3.6 Defects. - Bolts, nuts, and studs shall be free from seams, laps, burrs, loose scale, irregular surfaces and other defects which affect serviceability.

3.7 Workmanship. - Workmanship shall be compatible with the type and grade of the fastener and with the class of thread specified. Threads shall be free from slivers, splits, laps and other defects due to imperfections in the threading process.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. - For the purpose of inspection and tests a lot shall be homogeneous, that is consisting of bolts or nuts or studs of one material, type, grade and style, of the same nominal size and length, produced under essentially the same conditions and offered for inspection at one time. The lot shall consist of not more than the following number of bolts, or nuts or studs:

Sizes 1/4 to 1/2 inch - 20,000 pieces.
Sizes 9/16 to 1-1/2 inch - 10,000 pieces.
Sizes over 1-1/2 inch - 5,000 pieces.

4.2 Sampling. -

4.2.1 Acceptance quality level. - Unless otherwise specified in the contract or order, lot acceptance shall be based on an acceptance quality level (AQL) of 4.0 percent defective for visual and dimensional inspection. Determination of lot quality shall be by means of sampling inspection in accordance with Standard MIL-STD-105.

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4.2.2 Sampling for inspection. - From each lot the Government inspector shall select samples of random in accordance with Standard MIL-STD-105, using inspection level 1, single, double, or multiple sampling shall be used in accordance with the r[illegible]s specified herein. Lot acceptance or rejection shall be based on the requirements of the sampling plan used, based on the number of defective items found in the sample. A defective item is a bolt, nut or stud containing one or more defects.

(a) Small lots. - Single sampling only is permissible for lots containing not more than 180 pieces.

(b) Medium size lots. - Single or double sampling is permissible for lots containing 181 to 1300 pieces, inclusive.

(c) Large lots. - Single, double, or multiple sampling is permissible for lots containing over 1300 pieces.

4.2.3 Sampling for tests. -

4.2.3.1 Chemical analysis. - The Government inspector shall select a representative sample for chemical analysis from each lot. The sample shall consist of at least 4 ounces of clean fine drillings, millings or chips. Failure of the sample to comply with 4.6 shall be cause for rejection of the lot.

4.2.3.2 Mechanical and structural tests. - For purposes of mechanical and structural tests the lot (see 4.1) shall be weighed, or the weight shall be calculated. This weight in pounds shall be used as the lot size and a sample shall be selected in accordance with the appendix to Standard MIL-STD-105, using inspection level L-2. However, the sample selected shall be in pieces, in lieu of pounds, and the smallest sample size shall consist of two pieces. If any sample piece fails to conform to the mechanical and structural test requirements of this specification the lot shall be rejected.

4.2.3.3 Metallurgical tests. - For metallurgical tests the Government inspector shall select 2 samples at random from each lot. If one or both samples fail to meet the requirements specified in table II when tested in accordance with 4.5.1, the lot shall be rejected.

4.2.3.4 Protective coating tests. - When bolts, nuts and studs are electroplated, sampling shall be conducted in accordance with Specification QQ-P-416 or QQ-Z-325, as applicable. When hot dip zinc coatings are used, a minimum of four samples shall be selected from each lot (see 4.7).

4.3 Inspection. - Each of the samples selected in accordance with 4.2.2 shall be inspected for compliance to this specification. Visual inspection shall consist of an inspection for conformance with all requirements not requiring test or measurements. Inspection for dimensions shall consist of gaging or checking by suitable measuring instruments the threads and toleranced dimensions specified in tables VI through XVIII, as applicable (see 3.3 and 3.4).

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4.3.1 Gaging. - The contractor shall make available to the inspector the necessary gages and measuring instruments, the accuracy of which shall be certified in a manner acceptable to the inspector. In case of dispute, an independent laboratory acceptable to the inspector shall govern. For convenience, the toleranced dimensions which are normally inspected are summarized in table XIX.

Table XIX - Dimensions for which inspections may be made.

Description of dimension	Applicable fasterner	Method of checking
Pitch diameter	Bolts, studs, nuts	Thread gage
Major diameter	Bolts, studs	Ring gage
Minor diameter	Nuts	Plug gage
Height of head	Bolts	Micrometer or gage
Thickness of nut	Nuts	Micrometer or gage
Distance across flats	Bolts and nuts	Micrometer or gage
Width across corners	Bolts and nuts	Micrometer or gage
Total runout	Bolts	Gage
Body diameter	Bolts and studs	Gage and micrometer
Overall length	Bolts and studs	Scale
Thread length	Bolts and studs	Scale
Radius of fillet	Bolts	Template gage
Depth of square	Bolts	Micrometer
Width of square	Bolts	Micrometer
Diameter of head	Bolts	Micrometer
Width of slot	Nuts	Gage
Depth of slot	Nuts	Gage
Diameter of washer face	Bolts and nuts	Gage or micrometer
Straightness	Studs	Surface plate, and feeler gage

4.3.2 Rejected lots. - Prior to resubmitting a rejected lot for inspection the contractor shall rework or reinspect the lot for the deficiencies noted and remove all nonconforming items. The inspector shall select samples from the resubmitted lot in accordance with 4.2.2. The samples shall be inspected for the deficiency for which rejection was originally made.

4.4 Structure and mechanical tests. -

4.4.1 Each of the samples selected in accordance with 4.2.3.2 shall be tested in accordance with 4.4.2 through 4.4.10, as applicable.

4.4.2 Hardness test. -

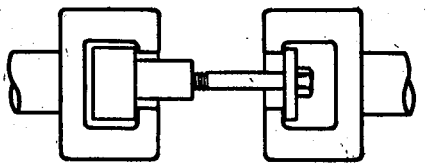
4.4.2.1 Bolts and studs. - The hardness of bolts and studs shall be measured on a transverse section through the threaded portion. The area on which the hardness measurement is made shall be located at a point 1/8 to 1/4 of the nominal diameter from the longitudinal axis on a properly prepared surface of the threaded end. For bolts and studs too long to be measured on the threaded end, a section at least one nominal diameter in length, shall be taken from the threaded end.

4.4.2.2 Nuts. - The hardness of nuts shall be measured on the end faces, the prongs of slotted nuts having been removed, when the Rockwell hardness tester is used. The Brinell hardness shall be taken on the side of the nut.

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4.4.3 Tensile test. -

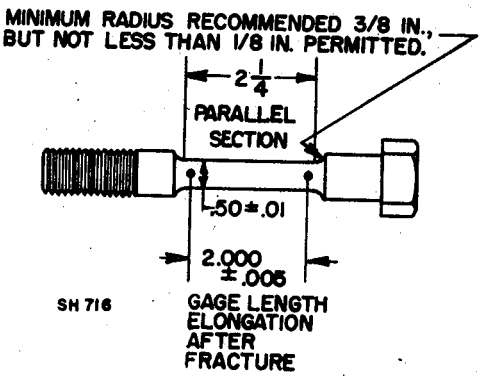
4.4.3.1 Arrangements. - Elasticproof and tension tests of bolts, and studs shall, preferably, be taken on the completed bolts, and studs with the load applied between the head and nut or suitable fixtures, either of which shall have sufficient thread engagement to develop the full strength of the bolts, and studs. The nut or fixture shall be assembled on the bolt or stud to the thread run out and then unscrewed three to six turns leaving six threads between the grips. The applicable tests shall be made in accordance with Specification QQ-M-151. The arrangement for testing a full size specimen is shown on figure 1.



SH 715

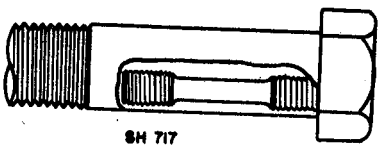
Figure 1 - Tension testing full-size bolt.

4.4.3.2 Turned and ground specimens. - When equipment of sufficient capacity is not available to test bolts, and studs in full size, tests shall be made as follows: Bolts, and studs 7/8 to 1-3/8 inches nominal size, inclusive, shall have their shanks concentric with the axis of the screw as shown on figure 2, leaving the head and the threaded portions intact. For bolts, and studs over 1-3/8 inch diameter, a type 1 specimen as described in Specification QQ-M-151 shall be turned such that the axis of the specimen is midway between the axis and outside surface of the bolt or stud (see figure 3).



SH 716

Figure 2 - Tension test specimens for bolt with turned down shank.



SH 717

Figure 3 - Location of standard round 2-inch gage length tension test specimen when turned from large-size bolt.

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4.4.3.3 Elastic proof load. - The overall length of a bolt or stud shall be measured at the true center line, and the head or body marked so the bolt or stud can be placed in the measuring fixture in the same position for subsequent measurement. The bolt or stud shall then be loaded to the applicable proof load specified in table I, or as specified in the applicable materials specification. The bolt or stud shall then be removed from the testing machine and its overall length measured. The length after loading shall not exceed the length before loading by more than 0.0005 inch (total).

4.4.3.3.1 The preferred method of measuring the length shall be between conical centers on the center line at each end of the bolt or stud using mating centers on the measuring anvils. Other recognized methods may be used, such as ball point anvils on prepared surfaces at the true center line.

4.4.3.4 Tensile strength and wedge test. - After the elastic proof load test specified in 4.4.3.3, for bolts types I, II and III, a 10 degree wedge shall be placed under the head or the nut and the tensile test continued to failure. The same sample may be used. A bolt shall meet the requirements of this test providing the following conditions are met:

(a) Tensile failure does not occur until the minimum tensile requirements specified in table I, of the applicable grade, or as specified in the applicable materials specification have been met.
(b) Tensile failure occurs in the threads or shank of the bolt.

Failure of the bolt at the juncture of the head and shank or fracture in the head shall be cause for rejection.

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4.4.3.4.1 Unit tensile strength. - The tensile strength value in pounds per square inch shall be based on the cross sectional area of the bolt where the break occurs. When the break is in the threads, the minimum breaking loads for steel bolts shall be as shown in tables XX and XXI as applicable.

Table XX - Minimum breaking loads of steel bolts, coarse-thread series.

Nominal size or basic major diameter of thread	Threads per inch and designation	Stress area of thread	Breaking load for steel of:				
			Grade 1	Grade 2	Grade 5	Grade 8	FS 303 cond. A
Inches		Square inches	Pounds	Pounds	Pounds	Pounds	Pounds
1/4	20 UNC	0.0318	1,750	2,190	3,820	4,770	2,380
5/16	18 UNC	.0524	2,880	3,620	6,290	7,860	3,930
3/8	16 UNC	.0775	4,260	5,350	9,300	11,620	5,810
7/16	14 UNC	.1063	5,850	7,330	12,760	15,940	7,970
1/2	13 UNC	.1419	7,800	9,790	17,030	21,300	10,640
9/16	12 UNC	.182	10,010	11,650	21,800	27,300	13,650
5/8	11 UNC	.226	12,430	14,460	27,100	33,900	16,950
3/4	10 UNC	.334	18,370	21,400	40,100	50,100	25,000
7/8	9 UNC	.462	25,400	25,400	53,100	69,300	34,600
1	8 UNC	.606	33,300	33,300	69,700	90,900	45,400
1-1/8	7 UNC	.763	42,000	42,000	80,100	114,000	57,200
1-1/4	7 UNC	.969	53,300	53,300	101,700	145,000	72,700
1-3/8	6 UNC	1.155	63,500	63,500	121,300	173,000	86,600
1-1/2	6 UNC	1.405	77,300	77,300	147,500	211,000	105,400
1-3/4	5 UNC	1.90	104,500	104,500	-------	-------	-------
2	4-1/2 UNC	2.50	137,500	137,500	-------	-------	-------
2-1/4	4-1/2 UNC	3.25	178,800	178,800	-------	-------	-------
2-1/2	4 UNC	4.00	220,000	220,000	-------	-------	-------
2-3/4	4 UNC	4.93	271,200	271,200	-------	-------	
3	4 UNC	5.97	328,000	328,000	-------	-------	-------
3-1/4	4 UNC	7.10	390,000	390,000	-------	-------	-------
3-1/2	4 UNC	8.33	458,000	458,000	-------	-------	-------
3-3/4	4 UNC	9.66	531,000	531,000	-------	-------	-------
4	4 UNC	11.08	609,000	609,000	-------	-------	-------

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Table XXI - Minimum breaking loads of steel bolts, fine-thread series.

Nominal size or basic major diameter of thread	Threads per inch and designation	Stress area of thread	Breaking load for steel of:				
			Grade 1	Grade 2	Grade 5	Grade 8	FS 303 cond. A
Inches		Square inches	Pounds	Pounds	Pounds	Pounds	Pounds
1/4	28 UNF	0.0364	2,000	2,510	4,370	5,460	2,730
5/16	24 UNF	.0580	3,190	4,000	6,960	8,700	4,350
3/8	24 UNF	.0878	4,830	6,060	10,540	13,170	6,580
7/16	20 UNF	.1187	6,530	8,190	14,240	17,800	8,900
1/2	20 UNF	.1599	8,790	11,030	19,190	24,000	11,990
9/16	18 UNF	.203	11,160	12,990	24,400	30,400	15,220
5/8	18 UNF	.256	14,100	16,380	30,700	38,400	19,200
3/4	16 UNF	.373	20,500	23,900	44,800	56,000	28,000
7/8	14 UNF	.509	28,000	28,000	58,500	76,400	38,200
1	12 UNF	.663	36,500	36,500	76,200	99,400	49,700
1-1/8	12 UNF	.856	47,100	47,100	89,900	128,400	64,200
1-1/4	12 UNF	1.073	59,000	59,000	112,700	161,000	80,500
1-3/8	12 UNF	1.315	72,300	72,300	138,100	197,000	98,600
1-1/2	12 UNF	1.581	87,000	87,000	166,000	237,200	118,600

4.4.3.4.2 Stripping of threads. - If failure occurs by stripping of the threads before reaching the minimum required tensile load the test shall be considered as inconclusive and another sample substituted.

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4.4.3.4.3 Ten degree wedge. - The 10 degree wedge for use in the tensile test shall be constructed as shown on figure 4.

CLEARANCE (C) SHALL BE APPROX. AS FOLLOWS:

NOM BOLT SIZE INCH	RADIUS INCH
1/4 TO 1/2	.030
9/16 TO 3/4	.050
7/8 TO 1 1/4	.063
1 3/8 TO 1 1/2	.094

MAX. RADIUS R, SHALL BE AS FOLLOWS:

NOM. BOLT SIZE INCH	RADIUS INCH
1/4 TO 1/2	1/32
9/16 TO 1	1/16
1 1/8 TO 1 1/2	1/8

c = CLEARANCE OF WEDGE HOLE.
d = NOMINAL DIAMETER OF BOLT.
R = MAX. RADIUS.
T = THICKNESS OF WEDGE AT SHORT SIDE OF HOLE EQUALS 1/2 (d).

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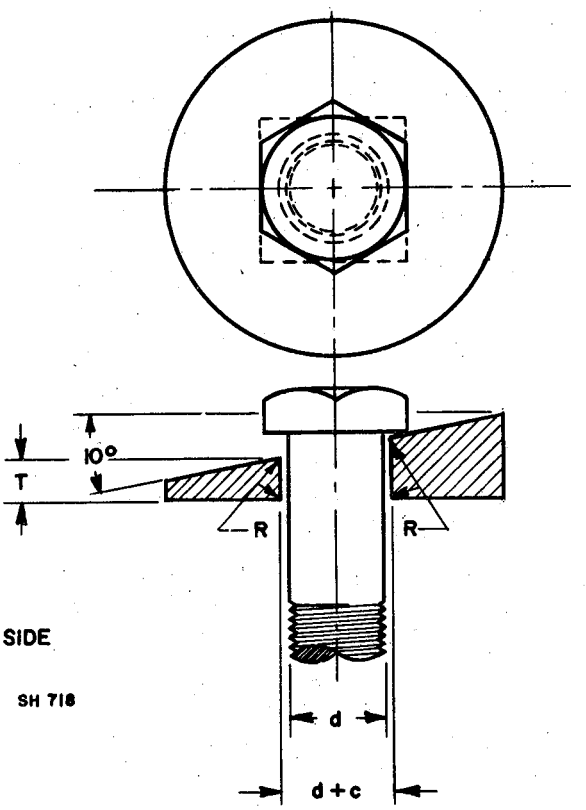


Figure 4 - Ten degree wedge test.

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4.4.4 Axially tensile strength. - When required the axially tensile test shall be made the same as the tensile strength and wedge test specified in 4.4.3 except the 10 degree wedge shall not be used. The same conditions of test as specified in 4.4.3.1 and 4.4.3.2 shall apply.

4.4.5 Yield strength. - When a determination of yield strength is required and when the bolt or stud is of sufficient size to permit use of the type I specimen of Specification QQ-M-151, a test bar shall be turned and tested, using an extensometer to measure deformation over the gage length. The yield strength shall be the load at which a permanent set in the gage length of 0.2 percent occurs.

4.4.6 Plastic deformation. - When required (see 6.2) and when tested in full size, the plastic deformation of bolts or studs shall be determined by the procedure in 4.4.5, except that the test shall be continued until at least 1/32 inch elongation is obtained on removal of the load. The bolt or stud shall not have ruptured in this test. For bolts not tested in full size (see 4.4.3) a standard turned test specimen shall be used; and when specified, the percent reduction of area and percent elongation in two inches shall also be determined.

4.4.7 Shear test. - When required (see 6.2), bolts, or studs shall be tested in shear to determine conformance to the specified requirements. The shear test shall be performed in a jig which is acceptable to the procuring agency. The jig shall subject the bolt, or stud to double shear transverse to the axis.

4.4.8 Bending test. - The bending test shall be made either by hammer blows or in a press or testing machine using suitable mandrels. For sizes to 1 inch, inclusive, the diameter of the mandrel shall be 1/2 the nominal diameter of the bolt or stud. For sizes over 1 inch the diameter of the mandrel shall be equal to the nominal diameter of the bolt or stud.

4.4.9 Flattening test. - The flattening test shall be made by hammer or press blows, reducing the original thickness by one half without visual evidence of cracks or rupture.

4.4.10 Stripping test. - The sample nut shall be assembled on a hardened threaded mandrel in a tension testing machine and the load specified in tables III or IV applied. Threads of the nut shall not strip at this load. If the threads of the mandrel are damaged during the test, the individual test shall be discarded. The mandrel shall be threaded to the unified or American National thread series class 3A except the major diameter shall be the minimum major diameter with a plus tolerance of 0.002 inch. The mandrel shall have a minimum hardness of Rockwell C45. In lieu of a hardened mandrel, a bolt conforming to these specifications may be used. If the unit tensile strength developed in the mandrel is required, the load obtained shall be divided by the stress areas given in tables III and IV as applicable.

4.5 Metallurgical test. -

4.5.1 Decarburization. - The threaded portion of bolts, nuts and studs selected in accordance with 4.2.3.3 shall be sectioned longitudinally through the axis and polished in such a manner that rounding of the thread edges is held to a minimum. The sections shall be etched with suitable reagents and examined with a metallurgical microscope. Measurements shall be made from the root of the thread to the beginning of the decarburized zone. Such distance shall be equal to or greater than that specified in table II for the applicable thread series. Measurements shall be made with a fixed scale micrometer eye piece or by throwing the image on the ground glass of the camera and making measures with a scale and dividers, or other suitable means.

4.6 Chemical analysis. - The sample selected in accordance with 4.2.3.1 shall be analyzed by a laboratory suitable to the Government inspector for compliance with the chemical requirements of table I, as applicable, or the applicable materials specification.

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4.7 Tests for protective coating thickness. -

4.7.1 Electroplated coatings. - The samples selected in 4.2.3.4 shall be tested for coating thickness in accordance with Specification QQ-P-416 or QQ-Z-325 as applicable.

4.7.2 Hot dip zinc coatings. - The samples selected in accordance with 4.2.3.3 shall be tested for conformance to 3.2.2.1 by means of the drop test specified in Specification MIL-Z-17871. Failure of the samples to conform to the requirements of 3.2.2.1 shall be cause for rejection of the lot.

4.8 Retests shall be granted as specified in Specification QQ-M-151.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. -

5.1.1.1 Preservation. - Uncoated ferrous bolts, nuts, and studs shall be cleaned by process C-1 of Specification MIL-P-116 and coated with preservative type grade 1 of Specification MIL-C-16173. Unless otherwise specified in the contract or order, (corrosion resisting steel or) nonferrous or zinc or cadmium coated bolts, nuts and studs shall not require a preservative coating.

5.1.1.2 Packaging. - Bolts, nuts, and studs, separated for item, type and size, shall be unit packaged in quantities specified (see 6.2). Unit containers shall conform to Specification PPP-B-566, PPP-B-676, MIL-B-4229, LLL-B-631, or LLL-B-636. The gross weight of folding cartons or set-up boxes shall not exceed 5-pounds; metal edge boxes shall not exceed 10 pounds. The weight of contents of fiberboard unit containers having a minimum Mullen test of 175 pounds shall not exceed 20 pounds; and of fiberboard containers having a minimum Mullen test of 200 pounds shall not exceed 40 pounds. Specified quantities in excess of 40 pounds shall be bulk packed directly in shipping containers specified in 5.2.2 or 5.2.3 as applicable.

5.1.2 Level C. - Bolts, nuts, studs, and raw stock shall be preserved and packaged in accordance with commercial practice.

5.2 Packing. -

5.2.1 Level A. - Bolts, nuts, and studs shall be packed in wood cleated plywood or nailed wood boxes conforming to Specification PPP-B-601 (overseas type) or JAN-P-106, respectively. Shipping containers shall be lined with a sealed waterproof case liner conforming to Specification MIL-L-10547. The gross weight of shipping containers shall not exceed approximately 200 pounds

5.2.2 Level B. - Bolts, nuts and studs shall be packed in cleated plywood or nailed wood boxes conforming to Specification PPP-B-601 (domestic type) or NN-B-621. The gross weight of shipping containers shall not exceed approximately 200 pounds.

5.2.3 Level C. - Bolts, nuts, and studs shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall conform to Consolidated Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order, marking shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Manufacturing data. - Threaded bolts, nuts and studs of carbon and alloy steels are generally produced as follows:

(a) Grade 2. - Bolts and studs are generally produced by cold heading or double extruding and stress relieving or by hot forging, and if necessary heat treated to meet requirements.

(b) Grade 5. - Bolts and studs are produced by hot or cold forging or machining from bar stock and heat treated to produce the required mechanical properties.

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(c) Grade 8. - Bolts and studs are made from fined grained medium carbon alloy steel quenched and tempered. The recommended minimum hardness in the as quenched condition is Rockwell C-47 for sizes to 1/2 inch inclusive and Rockwell C-45 for larger sizes. The micro structure should be free from ferrite.

(d) Nuts. - Nuts are made by cold punching or hot forging or machining from bar stock, and when necessary, heat treated to develop the required mechanical properties.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Name, type, style, and grade (see 1.2).
(c) Material (see 3.1). Condition of steel (see 3.1.2).
(d) Nominal size, and the lengths of bolts and studs (see 3.4)
(e) Thread series and class of fit (see 3.3).
(f) Quantity, expressed in terms of pieces of each item.
(g) The kind of protective coating when required (see 3.2)
(h) The finish, when other than unfinished studs are required (see 3.4.10).
(i) When case hardened steel fasteners are required.
(j) When shear strength and/or plastic deformation properties are to be determined (see 3.1.4.1.8 and 3.1.4.1.9).
(k) Whether level A, B, or C is required (see 5.1 and 5.2).
(ℓ) Quantity per unit package (see 5.1.2.2)

6.3 This specification does not cover all the types and sizes of the designated bolts, nuts and studs that are available. It is intended to cover only those sizes and types which are suitable for the majority of Bureau of Ships requirements.

6.4 Grades. - The grades of carbon and alloy steels specified in table I are in accordance with the recommended practice of the Society of Automotive Engineers.

6.5 Raw stock. - It is recommended, that stock for use by Bureau of Ships activities for the manufacture of bolts, nuts and studs, to meet the requirements of this specification, be procured in accordance with the following specifications:

Grade	Material specification
1	Commercial steel except Bessemer or resulfurized stock. Minimum tensile strength 55,000 p. s. i.
2	Federal Specification QQ-W-409, grade 2.
5	Federal Specification QQ-W-409, grade 5, except for usage at elevated temperatures 400 to 650°F. or for usage under shock and impact at temperatures below 30°F., the steel shall conform to Specification MIL-S-890 alloy #2.
8	Military Specification MIL-S-890 alloy #1.

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6.6 Unified standard nuts. - Nuts in sizes 1/4 to 5/8 inch inclusive in tables XIV and XVI have been unified dimensionally with British and Canadian standards. These nuts possess certain engineering and economic advantages over the same size nuts listed in tables XII, XIII and XV; hence, it is expected the nonunified nuts will become obsolete.

6.7 Cross reference. - The grades of steels and types of fasteners given in this specification which are equivalent to those given in the previous revision of Specification MIL-B-857 are as follows:

Grades and classes of steel.

This specification	MIL-B-857
Grades	Classes
1	C
2	B
5	A
8	-

Types of bolts.

This specification	MIL-B-857 (1.2)
Types	
I	None
II	Type B semifinished
III	Type B finished
IV	Type C
V	Type D
None	Type A

Types of nuts

This specification	MIL-B-857 (1.2)
Types	
I	Type A unfinished hexagon
II	Type A semifinished hexagon
III	Type A finished hexagon
IV	Type A semifinished hexagon jam
V	Type A finished hexagon jam
VI	Type A semifinished hexagon slotted
VII	None
None	Type B square, unfinished

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6.8 Resume of test requirements. - For convenience a summary is given of the mechanical and structural tests to which fasteners may be subjected.

Type of test	Application
Elastic proof load and tensile and 10° wedge test (see 4.4.3)	Bolts, types I, II and III made of carbon and alloy steels
Elastic proof load and axial tensile test (see 4.4.3.3 and 4.4.4)	Bolts types IV and V and studs made of carbon and alloy steels
Yield strength and tensile and 10° wedge test (see 4.4.3 and 4.4.5)	Bolts, types I, II and III made of corrosion resistance steel and nonferrous metals
Elastic proof load and axial tensile test (see 4.4.3.3 and 4.4.4)	Bolts types IV and V and studs made of carbon and alloy steels
Hardness test (see 4.4.2)	Bolts and studs when too short for tensile tests, and nuts, made of carbon and alloy steels
Bend test (see 4.4.8)	Bolts and studs of carbon and alloy steels when of sufficient length
Flattening test (see 4.4.9)	All nuts, and carbon and alloy bolts and studs too short for bend tests
Strip test (see 4.4.10)	Nuts
Shear test (see 4.4.7)	Bolts and studs when specified
Plastic deformation (see 4.4.6)	Bolts and studs when specified

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APPENDIX I

SCREW THREADS STRESS AREA

10. The stress area used for determining the strength of bolts and studs in tension corresponds to a transverse section midway between the pitch and minor diameters of the thread. The formula used for computing this area is:

$$3.1416 \left[\frac{E}{2} - \frac{3H}{16}\right]^2$$

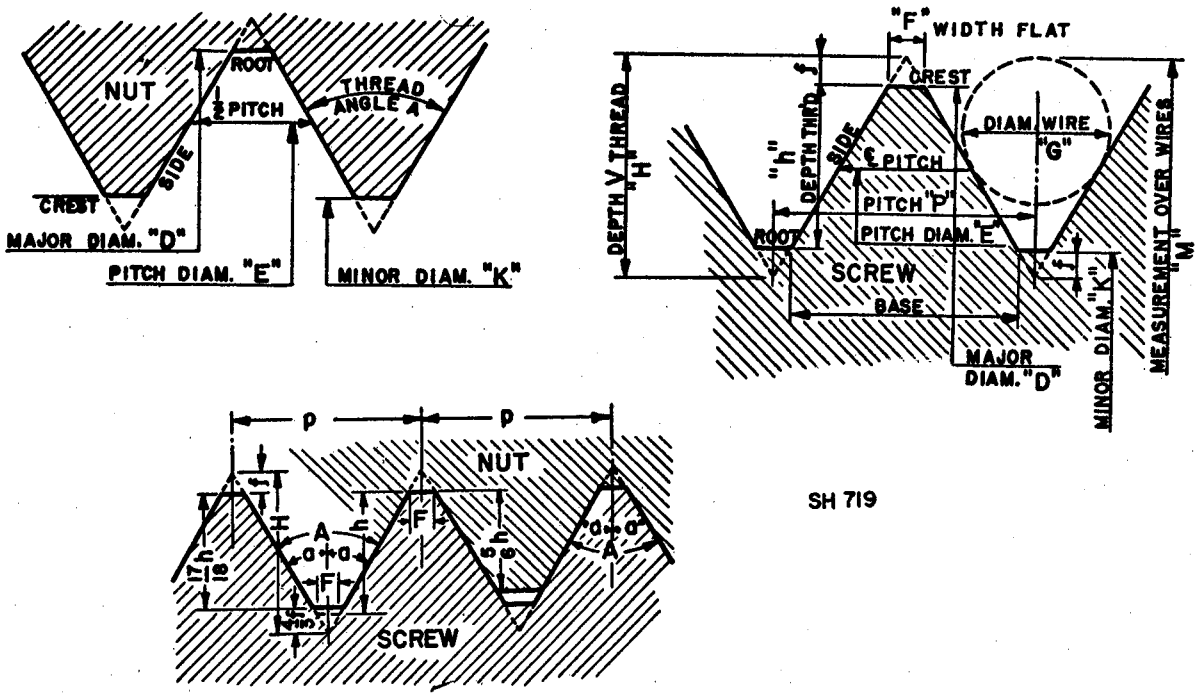
Where:
E is the basic pitch diameter and
H is the depth of the sharp V-thread.

10.1 Dimensions used for calculating stress areas shall be as specified in table XXII (see figs. 5 and 6).

Table XXII - Dimensions used for calculating stress areas.

Nominal size	UNC - thread			UNF - thread		
	T.p.i.	E	H	T.p.i.	E	H
1/4	20	0.2175	0.04330	28	0.2268	0.03093
5/16	18	.2764	.04811	24	.2854	.03608
3/8	16	.3344	.05413	24	.3479	.03608
7/16	14	.3911	.06186	20	.4050	.04330
1/2	13	.4500	.06662	20	.4675	.04330
9/16	12	.5084	.07217	18	.5264	.04811
5/8	11	.5660	.07873	18	.5889	.04811
3/4	10	.6850	.08660	16	.7094	.05413
7/8	9	.8028	.09623	14	.8286	.06186
1	8	.9188	.10825	12	.9459	.07217
1-1/8	7	1.0322	.12372	12	1.0709	.07217
1-1/4	7	1.1572	.12372	12	1.1959	.07217
1-3/8	6	1.2667	.14434	12	1.3209	.07217
1-1/2	6	1.3917	.14434	12	1.4459	.07217
1-3/4	5	1.6201	.17321	--	------	------
2	4-1/2	1.8557	.19245	--	------	------
2-1/4	4-1/2	2.1057	.19245	--	------	------
2-1/2	4	2.3376	.21651	--	------	------
2-3/4	4	2.5876	.21651	--	------	------

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A = 60 degrees.
a = 30 degrees.
n = Number of threads per inch.
p = Pitch of thread.
H = 0.866025p (depth of 60 degrees sharp V thread).
h = 0.649519p (depth of American National form thread).
5/8h = 0.541266p (maximum depth of engagement).
17/18h = 0.613435p.
F = 0.125000p (width of flat at crest and root of American National form).
f = 0.108253p
= 1/8H
= 1/6h
} Depth of truncation.

Figure 5 - American National form of thread notation.

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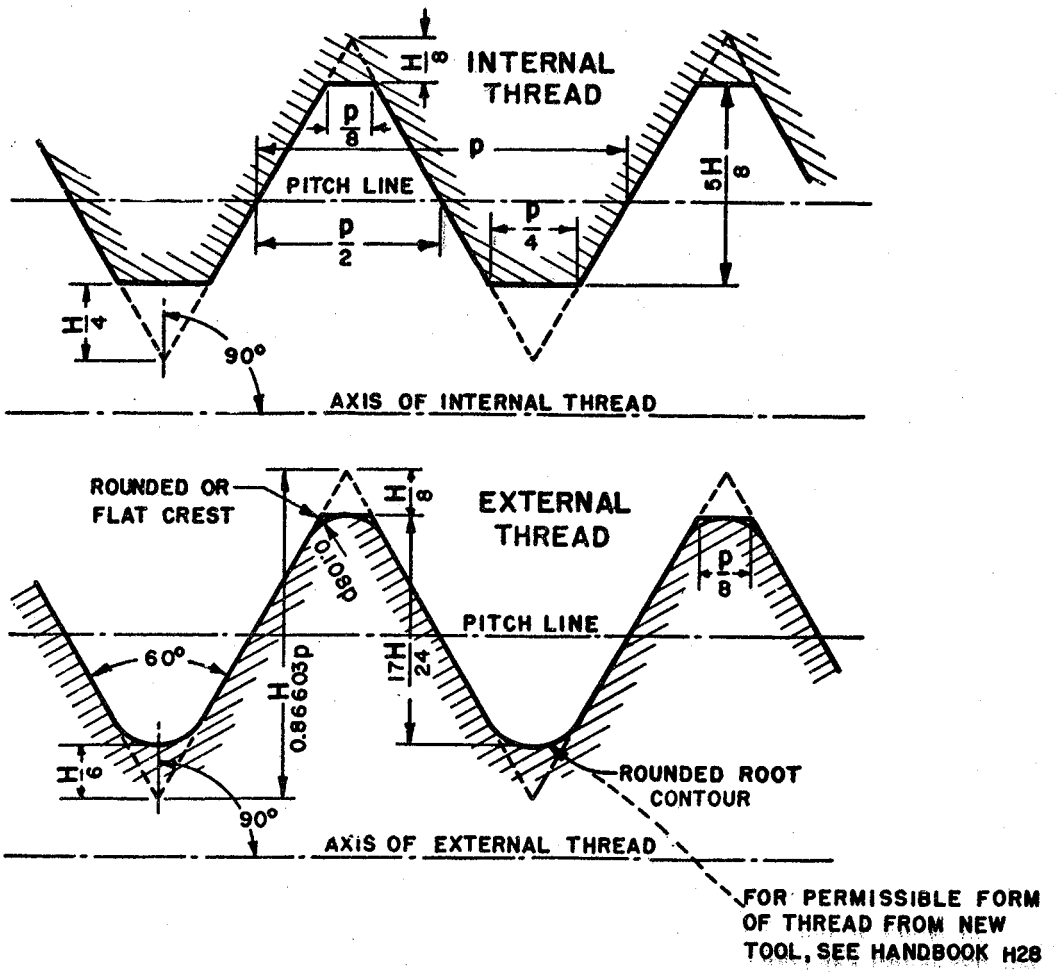


Figure 6 - Unified internal and external screw thread forms.

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APPENDIX II

FORMULAS

10. Formulas for bolt heads shall be as specified in table XXIII and for nuts as specified in table XXIV

Table XXIII - Formulas for bolt heads.

Product	Width across flats			Height of head		
	Basic[1]		Tolerance (minus)	Nominal[2]		Tolerance (plus or minus)
	Size	Width		Size	Height	
Regular hexagon bolt	1/4 5/16 to 4	1-1/2D + 1/16 1-1/2D	0.050D	1/4 to 7/16 1/2 to 7/8 1 to 1-7/8 2 to 3-3/4 4	5/8D + 1/64 5/8D + 1/32 5/8D + 1/16 5/8D + 1/8 5/8D + 3/16	Plus tolerance only 0.016D + 0.012 Minus tolerance adjusted so that minimum head height is same as minimum head height of regular semi-finished hexagon bolt
Regular semifinished hexagon bolt	1/4 5/16 to 4	1-1/2D + 1/16 1-1/2D	0.050D	1/4 to 7/8 1 to 1-7/8 2 to 2-3/4 3 3-1/4 to 3-3/4 4	5/8D 5/8D - 1/64 5/8D - 1/32 5/8D 5/8D - 1/16 5/8D	Size: 1/4 to 1; 1-1/8 to 4 Tolerance: 0.015D + 0.003; .016D ± .012
Finished hexagon bolt Hexagon head cap screw	1/4 5/16 to 3	1-1/2D + 1/16 1-1/2D	Size: 1/4 to 5/8; 1/4 to 1; 1-1/8 to 3 Tolerance: 0.015D + 0.006; .025D + .006; .050D (NOTE: For finished hexagon bolts in sizes 1/4 to 1 inch tolerance may be 0.050D when product is hot made.)	Same as for regular Semifinished hexagon bolt		Same as for regular Semifinished hexagon bolt

All dimensions given in inches.
[1] Adjusted to sixteenths.
[2] 1/4 to 1 inch sizes adjusted to sixty-fourths.
1-1/8 to 2-1/2 inch sizes adjusted upward to thirty-seconds.
2-3/4 to 4 inch sizes adjusted upward to sixteenths.
D = Nominal bolt size.
W = Width across flats.

Width across corners

Hexagon

For all hexagon bolt heads, max. width across corners equals 1.1547 x W (max.) and min. width across corners equals 1.14 x W (min.).

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Table XXIV - Formulas for nuts.

Type of nut	Width across flats				Thickness of nut			
	Basic[1]		Tolerance (minus)		Nominal[2]		Tolerance (plus or minus)	
	Size	Width	Size	Tolerance	Size	Thickness	Size	Tolerance
Finished hexagon	1/4	1-1/2D + 1/16	1/4 to 5/8	0.015D + 0.006	1/4 to 5/8	7/8D	1/4 to 5/8	0.015D + 0.003
Finished hexagon slotted	5/16 to 3	1-1/2D	3/4 to 3	.050D	3/4 to 1-1/8	7/8D - 1/64	3/4 to 3	.016D + .012
					1-1/4 to 2	7/8D - 1/32		
					2-1/4 to 3	7/8D - 3/64		
Finished hexagon-jam	1/4	1-1/2D + 1/16	1/4 to 5/8	.015D + .006	1/4 to 5/8	(See table)		
	5/16 to 3	1-1/2D	3/4 to 3	.050D	3/4 to 1-1/8	1/2D + 3/64	1/4 to 5/8	.015D + .003
					1-1/4 to 2	1/2D + 3/32	3/4 to 3	.016D + .012
					2-1/4	1/2D + 5/64		
					2-1/2 to 3	1/2D + 13/64		
Regular hexagon	1/4 to 5/8	1-1/2D + 1/16		0.050D		7/8D		0.016D + 0.012
	3/4 to 3	1-1/2D						
Regular hexagon-jam	1/4 to 5/8	1-1/2D + 1/16		.050D	1/4 to 7/16	1/2D + 1/32		.016D + .012
	3/4 to 3	1-1/2D			1/2 to 1-1/8	1/2D + 1/16		
					1-1/4 to 2-1/4	1/2D + 1/8		
					2-1/2 to 3	1/2D + 1/4		
Regular semifinished hexagon	1/4 to 5/8	1-1/2D + 1/16		.050D	1/4 to 1-1/8	7/8D - 1/64		.016D + .012
Regular semifinished hexagon slotted	3/4 to 3	1-1/2D			1-1/4 to 2	7/8D - 1/32		
					2-1/4 to 3	7/8D - 3/64		
Regular semifinished hexagon-jam	1/4 to 5/8	1-1/2D + 1/16		.050D	1/4 to 7/16	1/2D + 1/64		.016D + .012
	3/4 to 3	1-1/2D			1/2 to 1-1/8	1/2D + 3/64		
					1-1/4 to 2	1/2D + 3/32		
					2-1/4	1/2D + 5/64		
					2-1/2 to 3	1/2D + 13/64		

All dimensions given in inches.
[1]Adjusted to sixteenths
[2]Adjusted to fractions.
D = Nominal nut size.
W = Width across flats.

Width across corners

Hexagon

For all hexagon nuts, max. width across corners equals 1.1547 x W (max.) and min. width across corners equals 1.14 x W (min.).

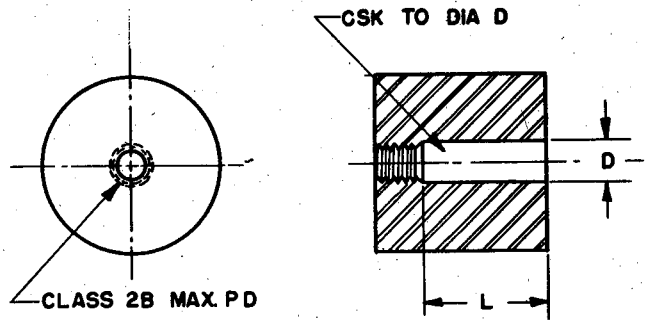
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APPENDIX III

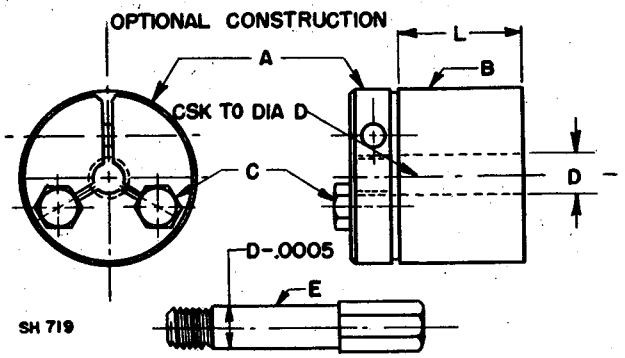
THREAD RUNOUT SLEEVE GAGE

10. Sleeve gages capable of checking the thread eccentricity and bow of shank on finished hexagon bolts are shown on figure 7. The optional construction permits the use of different length sleeves to accommodate different bolt lengths.

10.1 Ring gage A in the optional construction is centered in position on sleeve B by means of positioning plug E and is secured in position by means of attachment screws C. The ring gage is also set to maximum P.D. by positioning plug E. The internal diameter D of the sleeve equals the nominal diameter of the bolt plus runout allowance. The sleeve should extend beyond the last thread of the bolt to be inspected but should not exceed 3 inches in length. Failure of the bolt to enter the ring gage or interference between the sleeve and bolt while engaging the ring gage indicates excessive thread runout.



D - MAX. DIA OF SCREW PLUS CLEARANCE ALLOWANCE.
L - LENGTH OF SCREW MINUS ONE SCREW DIAMETER.
SHALL BE HARDENED



A- RING GAGE SET TO CLASS 2B MAX. PITCH DIA.
B- SLEEVE.
C- ATTACHMENT SCREWS.
D- DIA OF HOLE IN SLEEVE - MAX. SCREW DIA PLUS CLEARANCE ALLOWANCE.
E- POSITIONING AND SETTING PLUG, THREADS CLASS 2B MAX. P.D., FOR ADJUSTING RING GAGE AND CENTERING SLEEVE.
L- LENGTH OF SLEEVE - LENGTH OF SCREW MINUS ONE SCREW DIAMETER.

Figure 7 - Thread runout sleeve gage.

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Tables VI, VII, VIII, IX, X, XII, XIII, XIV, XV, XVI, Appendix II (tables XXIII and XXIV) and Appendix III, pages 12, 13, 14, 15, 16, 19, 20, 22, 24, 26, 48, 49 and 50, respectively.

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